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Abstract: Retroflexion in South Asia has been the subject of at least two previous typological studies: Ramanujan and Masica (1969) and Tikkanen (1999). Despite their many virtues, these studies are limited by the size of their data samples, their dependence on qualitative data without quantitative analysis, and their use of hand-drawn maps. This paper presents the results of an entirely new survey of retroflexion in South Asia – one that incorporates a larger language sample, quantitative analysis, and computer-generated maps. The study focuses on the genetic and geographic distribution of various retroflex subsystems, including retroflex obstruents, nasals, liquids, approximants and vowels. While it is possible to establish broad statistical correlations between specific types of contrast and individual language families (or sub-families), the study finds that the distribution of most retroflex systems is more geographic in nature than genetic. Thus, while retroflexion is characteristic of South Asia as a whole, each type of retroflex system tends to cut across genetic lines, marking out its own space within the broader linguistic area.

Keywords: phonology; retroflex; typology; areal feature

Retroflexion in South Asia: Typological, genetic, and areal patterns

1. Introduction

In a landmark study, Ramanujan and Masica (1969) sketched the first large-scale phonological typology of South Asia. Their approach was both typological and dialectological, identifying patterns of variation across phoneme inventories and mapping those patterns as isoglosses on a map of South Asia. Among other things, they examined the typology of retroflex consonant systems and their distribution. The authors viewed their study as “more in the nature of a preliminary report than a conclusive statement” (544), and anticipated its “further refinement into a definitive typological inventory of South Asian phonologies by the cooperative endeavor of all concerned” (552). Thirty years later, Tikkanen (1999) presented a revised typology of retroflexion in South Asia, along with a map showing the distribution of the various types that he identified (cf. Pärpola 1994: 166).¹

Despite their many virtues, Ramanujan and Masica’s (1969) original study and Tikkanen’s (1999) contribution have their limitations. First of all, the original study was limited to data that was available in the late 1960s, and there are notable gaps. For instance, there are very few Tibeto-Burman languages in the survey, and none whatsoever from northwest India or Nepal. Tikkanen (1999) made significant improvements in this area, but given the current state of language documentation and description in South Asia, there is still room for further improvement (see Table 1). Secondly, both studies are purely qualitative, not quantitative. They support their observations by listing examples but do not provide statistics to indicate the relative frequency of phonological patterns. Finally, the maps in both studies are hand-drawn and in some cases cluttered with a lot of information. As a result, they can be difficult to interpret and areal patterns do not always stand out clearly.

The present paper takes a fresh look at the question of retroflexion in South Asia. Like the earlier studies, it is concerned with both the typology of retroflex phoneme inventories and their genetic and geographic distribution in South Asia. The current study can be viewed as a revision of earlier observations on the topic, offered in the spirit of “further refinement” anticipated by Ramanujan and Masica (1969). To begin with, it is based on a much larger language sample that incorporates data from the wealth of descriptive studies published in recent decades. This includes a substantial number of Tibeto-Burman languages from northwest India, Nepal, and elsewhere.

¹ Other typological studies have focused on specific sub-regions, such as Nepal (Michailovsky 1988) or Northeast India (Neukom 1999); focused on phonetics as opposed to phonology (Ramaswami 1999); or covered much of the same ground as Ramanujan and Masica’s (1969) original study (Reddy 2003).

Secondly, the study quantifies the frequency of various retroflex phonemes and inventory types, using simple statistics. Finally, it provides improved, computer-generated maps, illustrating the genetic and geographic distribution of typological patterns.

The results of the study corroborate many of the key observations in Ramanujan and Masica (1969) and Tikkanen (1999), while at the same time refining them in important ways. It affirms that retroflex segments of one kind or another occur in the vast majority of South Asian languages, including some from each of the main families represented in the region, and that the distribution of languages with retroflexion corresponds very closely to the area of South Asia. However, it also highlights the fact that retroflexion extends well beyond the limits of South Asia into what is commonly considered East Asia (i.e., China), a detail that is not given much attention elsewhere. The study also examines the considerable typological variation in retroflex segment inventories within South Asia. While it is possible to establish broad statistical correlations between specific types of contrast and individual language families (or sub-families), the study finds that the distribution of most retroflex systems is more geographic in nature than genetic. Each type tends to cut across genetic boundaries to some degree, marking out its own geographic space within the broader linguistic area.

The paper is organized as follows. Section 2 reviews some background information on the study. Section 3 provides a broad overview of retroflexion in South Asia and environs, irrespective of manner of articulation. The rest of the paper focuses on the genetic and geographic distribution of specific retroflex subsystems, beginning with obstruent systems in section 4, and continuing with nasal systems in section 5, liquid systems in section 6, and retroflex approximants and vowels in section 7. Finally, some concluding remarks are offered in section 8.

2. Methodology

The current study is based on a survey of phonological literature describing 260 language varieties from South Asia and surrounding regions. Before examining the results of this survey, it will be useful to review some details concerning the methodology employed. The following subsections review details concerning the languages of South Asia, the sample employed for the survey, the mapping of phonological traits, and the nature of those traits.

2.1. *Languages of South Asia*

For the purpose of the present study, South Asia is defined as the area comprising the states of India, Pakistan, Nepal, Bhutan, Bangladesh, Sri Lanka and Maldives.² Four main language families are represented in this area: Indo-Iranian (a sub-branch of Indo-European), Dravidian, Austro-Asiatic, and Tibeto-Burman (a sub-branch of Sino-Tibetan). The Andamanese languages also fall within the borders of South Asia, along with a few Tai-Kadai languages and isolates. Throughout the paper, these minor families and isolates are lumped together under the category of ‘other’ languages.

The geographic distribution of these families is shown in Map 1. Indo-Iranian languages dominate the northern and western parts of South Asia. Most languages of this family belong to the Indo-Aryan branch, with languages of the Iranian branch limited to western Pakistan and adjacent areas beyond its borders. In Afghanistan, the small group of Nuristani languages is considered a third branch of Indo-Iranian. The Dravidian family dominates the south and has no members outside of South Asia. Most Austro-Asiatic languages of South Asia belong to the Munda branch, which is concentrated in eastern India. Apart from a few Khasian languages in northeast India and the Nicobarese languages of the Nicobar Islands, all other Austro-Asiatic branches are concentrated in Southeast Asia. The Tibeto-Burman family is spread across the Himalayas on the northern and northeastern peripheries of South Asia and adjacent parts of China and Southeast Asia. Speakers of Indo-Iranian and Dravidian languages account for about 78% and 20% of the South Asian population, respectively, whereas Austro-Asiatic and Tibeto-Burman are limited to minority groups within the region (Ebert 2006).

2.2. *Language samples*

The language sample used for this study is primarily a convenience sample based on availability of data. However, every effort was made to ensure that all language families and geographic regions were well represented. Table 1 shows a breakdown of the sample by language family and (where appropriate) sub-family. The ‘Current Maps’ column represents language varieties included on maps in the current study, whereas the ‘Current Stats’ column represents those used to calculate statistics. A total of 260 language varieties is included on the maps, but statistics were computed on a subset of 205 varieties that fall within the boundaries of South Asia (as defined above). Thus, the statistics reflect facts about South Asian languages, while the maps include data

² Some definitions of South Asia may include extensions into Afghanistan, the Tibetan region of China, and even Myanmar (Masica 1992). More commonly, these countries are classified as Central, East, and Southeast Asia, respectively, although parts of them may constitute transitional zones between those regions and South Asia (e.g., Tikkanen 2008).

from surrounding regions in order to provide context and avoid begging the question of a South Asian linguistic area. The columns labelled ‘R&M 1969’ and ‘Tikk. 1999’ represent the samples used in Ramanujan and Masica (1969) and Tikkanen (1999), respectively. They are included here for comparison. With the exception of Iranian, the current study incorporates an equivalent or larger sample for every family and sub-group, compared to the previous studies.

Table 1: Language samples

Classification	Sub-Group	R&M 1969	Tikk. 1999	Current Stats	Current Maps
Indo-Iranian	Iranian	4	14	5	13
	Nuristani	1	4	0	4
	Indo-Aryan	28	39	64	64
Dravidian	—	15	20	37	37
Austro-Asiatic	Munda	7	7	15	15
	Khasian	1	1	2	2
	Nicobarese	1	1	2	2
	Other AA	1	0	0	4
Sino-Tibetan	Tibeto-Burman	8	30	72	96
	Chinese	0	1	0	2
Other	Isolates	1	1	2	2
	Tai-Kadai	0	0	3	5
	Andamanese	0	0	3	3
	Afro-Asiatic	0	0	0	6
	Turkic	0	3	0	4
	Mongolic	0	1	0	1
Totals		67	122	205	260

Statistics concerning South Asia as a whole are computed on a sample of 205 language varieties. This sample includes family units such as Iranian, Khasian, Nicobarese, and the various minor families in the ‘other’ category. These family units have very few members within the region. For this reason, the study makes no attempt to characterize them statistically. Instead, statistical comparisons of family units within South Asia are limited to the four units that constitute the majority of South Asian languages: Indo-Aryan, Dravidian, Munda, and Tibeto-Burman. Together, these account for 188 (92%) of the 205 language varieties in the South Asian sample.

2.3. Maps

All maps presented in this paper were generated using a Geographic Information System (GIS) software called Quantum GIS (QGIS). For convenience, language locations were plotted as points on the map as opposed to areas, a practice that is

commonly employed in typological surveys of this kind (cf. Dryer and Haspelmath 2013).³ In most cases, language locations were determined based on their data source. For instance, if the source study indicated that data was collected from a particular locality, then coordinates for that locality were used to situate the language on the map. Where such details were not provided, or in the case of certain major languages spoken over larger areas, an effort was made to situate languages near the centre of the area over which they are spoken. For this purpose, maps from the *Ethnologue* (Lewis et. al. 2016) were consulted, and in some cases coordinates from the *World Atlas of Language Structures* (Dryer and Haspelmath 2013) were used.

On each map, languages are represented by small circular icons. The genetic affiliation of each language is indicated by an abbreviation within the circles: IA for Indo-Aryan, D for Dravidian, M for Munda, T for Tibeto-Burman, etc. In addition, textured shading is used to highlight the geographic distribution of phonological traits. The genetic distribution of phonological traits can be observed by examining the affiliation of languages that fall within any given shaded area. In this way, the maps illustrate both the genetic and geographic distribution of traits simultaneously.

In order to compute the boundaries of shaded areas, binary numeric values were assigned to represent the presence or absence of a trait in each language (e.g., 1 = present, 0 = absent). Shading was then generated using the Inverse Distance Weighting interpolation method in QGIS. This function uses the known values of points plotted on the map to estimate the unknown values of intervening spaces. Application of this function yielded a raster layer with values ranging from 0.0 to 1.0. A contour was extracted from this layer at a value of 0.5, which represents the halfway point between presence (1.0) and absence (0.0) of a feature. This contour was then taken as the feature boundary. It is important to recognize that the shaded areas generated by this method cannot be taken as precise isoglosses for any given feature. They are approximate at best and are included primarily as a visual aid to highlight areal patterns.

2.4. *Phonological traits*

The study is concerned primarily with phonological oppositions, not their phonetic implementation. This means that segments or contrasts that are given different phonetic descriptions in the literature are sometimes treated as equivalent from an abstract phonological point of view. For instance, a distinction is sometimes made between two degrees of retroflexion: a weaker form of apical alveolar or post-alveolar retroflexion and a stronger form of sub-apical palatal retroflexion. The former is said

³ The kind of data required to map languages as areas is not available in most cases. Moreover, demarcating boundaries between languages is complicated by the fact that they regularly overlap.

to be characteristic of many Indo-Aryan and Tibeto-Burman languages, while the latter is associated with Dravidian languages (Ladefoged and Bhaskararao 1983). However, phonetic studies reveal a wide range of variation in languages of both types, depending on the speaker, speech rate, manner of articulation, and vowel context (Reddy 1986; Dixit and Flege 1991; Dart and Nihalani 1999; Khatiwada 2007). Moreover, the distinction is rarely significant from a phonological point of view.⁴ Thus, the two types are not distinguished in the present study; both are treated as retroflex. All consonants counted as retroflex stand in phonological opposition to some dental or denti-alveolar counterpart with a comparable manner of articulation.

Segments that are limited to loanwords were counted only if they were described as frequent and well-integrated in colloquial speech. Otherwise, they were excluded. With these points in mind, we can turn to the survey of retroflexion in South Asia.

3. Retroflexion as an areal feature of South Asia

South Asia is widely recognized as a linguistic area – a geographic region in which languages of different genetic stock have come to resemble one another through a history of contact and convergence (Emeneau 1956; Masica 1976; Ebert 2006). Retroflexion is identified as a defining feature of the area (Kuiper 1967; Ramanujan and Masica 1969; Bhat 1973). Most languages of South Asia have retroflex segments of one kind or another, regardless of their genetic affiliation. This section surveys the full extent of retroflexion in South Asia and neighbouring regions, independent of manner of articulation.

Figure 1 shows the frequency of retroflexion in the South Asian sample compared to that of the *UCLA Phonological Segment Inventory Database* (UPSID), which represents a genetically and geographically balanced sample of 451 languages of the world (Maddieson 1984; Maddieson & Precoda 1990). Retroflexion is a relatively marked feature, cross-linguistically. Only about 20% of languages in the UPSID database have retroflex segments of some kind; while 80% do not. In South Asia, these proportions are inverted; only 22% of languages lack retroflexion, while 78% have retroflex segments of one kind or another.

Retroflexion is most prevalent within the Dravidian, Indo-Aryan, and Munda families, and least prevalent within the Tibeto-Burman family, as shown in Figure 2. All Dravidian languages have retroflexion. The Indo-Aryan family is not far behind, with retroflexion in all members except for Assamese. Recall that Dravidian and Indo-

⁴ To be sure, a few South Dravidian languages contrast apical alveolar and retroflex consonants. However, this contrast always occurs in addition to the more basic distinction between lamino-dental and apical articulations. This basic contrast is the focus of the present study, and languages that exhibit no further contrast are expected to show more variation in phonetic implementation.

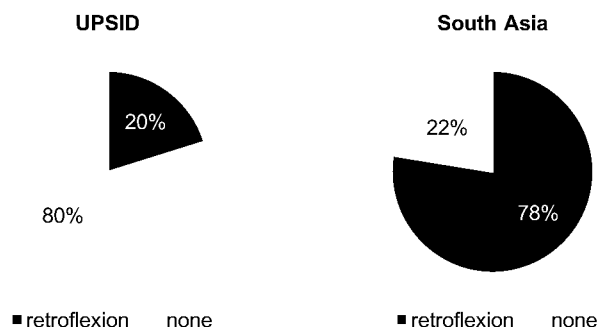


Figure 1: Proportion of languages with and without retroflexion

Iranian languages together account for as much as 98% of the population. Thus, the vast majority of the population speaks a language with retroflexion. Retroflexion is also reported in all Munda languages, which constitute the majority of Austro-Asiatic languages in the region.⁵ It does not occur in any other Austro-Asiatic language of South Asia, with the possible exception of Pnar in northeast India.⁶ Only about half of all Tibeto-Burman and ‘other’ languages of South Asia have retroflexion of some kind. Within the ‘other’ category, retroflexion occurs in Burushaski (isolate) and the Andamanese languages, but not in Kusunda (isolate) or the Tai-Kadai languages.

The distribution of retroflexion within Tibeto-Burman warrants further comment. As shown in Figure 3, retroflexion occurs primarily in the Western Tibeto-Burman branch, most notably in the Bodish group (e.g., Ladakhi, Gurung, Tamang, Sherpa, Dzongkha, Tshangla, Tibetan, etc.), and Western Kiranti (e.g., Sunwar, Jero, Thulung, Wambule, etc.). It is also characteristic of Qiangic and rGyalrongic languages in China. Retroflexion does not occur in any Eastern Kiranti language surveyed here, and is rare in other sub-groups, including Central Tibeto-Burman and the so-called ‘Sal’ languages, most of which are concentrated in Northeast India and adjacent parts of China and Southeast Asia.⁷

⁵ Previous studies list Korku and Sora as Munda languages without retroflexion (Ramanujan and Masica 1969; Masica 1992; Neukom 1999). These studies cite Zide (1960) for Korku, and Stampe (1965) for Sora. While these sources report no retroflex contrast among stops, they do report a contrast between /r/ and /ɾ/. Thus, Korku and Sora are treated here as having retroflexion in the class of liquids.

⁶ Pnar has a contrast between lamino-dental and apico-alveolar stops that might be construed as a weak form of retroflexion (Ring 2015). I have interpreted it as such for the present study.

⁷ The statistics in Figure 3 are based on the total sample of 96 Tibeto-Burman languages, not just those within the core South Asian countries. The classification scheme is based on the nineteenth edition of

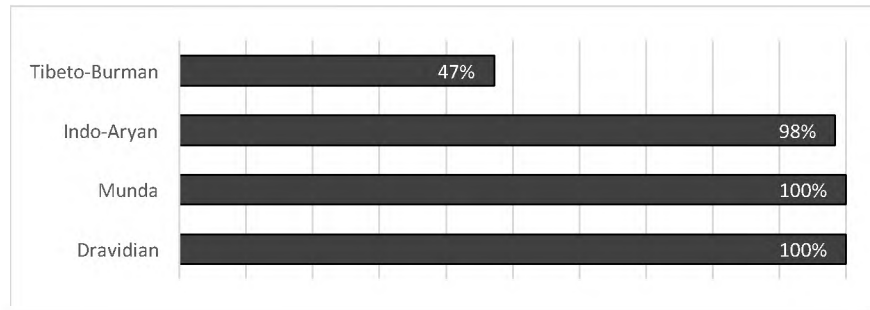


Figure 2: Percentage of South Asian languages with retroflexion by language family

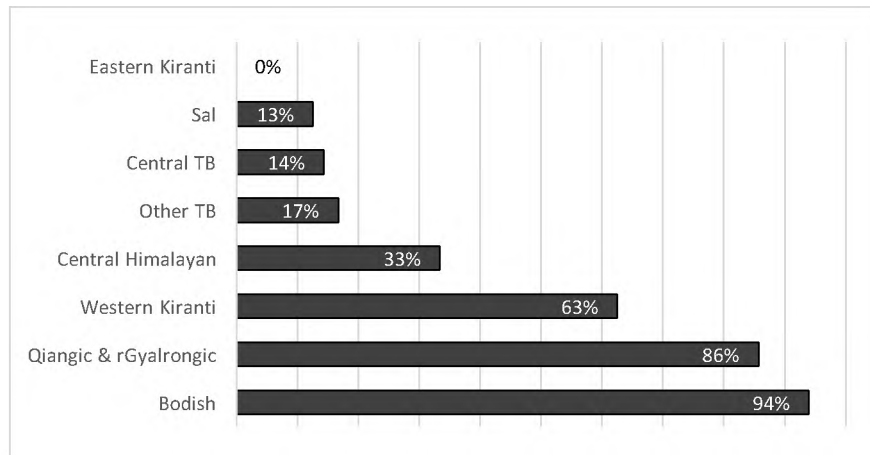


Figure 3: Percentage of languages with retroflexion in selected Tibeto-Burman subgroups.

Map 2 affirms that the main concentration of languages with retroflexion corresponds very closely to the area of South Asia (i.e., the Indo-Pakistani subcontinent). Within South Asia, retroflexion is widespread from the south to the northwest, but fades out in Nepal and northeast India. Significantly, the retroflex area extends beyond the northeast borders of South Asia, well into China (i.e., East Asia). This extension affects a great many Tibeto-Burman languages of China, including many outside of the Tibet Autonomous Region, and reaches as far east as Mandarin (off the map), which belongs to the Chinese branch of Sino-Tibetan. This point is worth stressing because it is not clearly addressed by Ramanujan and Masica (1969) or

the *Ethnologue* (Lewis et. al. 2016). The ‘other TB’ category consists of a few Ngwi-Burmese and Karenic languages.

Tikkanen (1999), and is rarely acknowledged or discussed in the South Asian literature. As we will see, however, the retroflex consonant systems in Sino-Tibetan languages of China are typologically distinct from those commonly found in South Asia.

A closer look at Nepal and northeast India is provided in Map 3. A few observations can be made. First of all, just like areas *with* retroflexion, areas *without* it also cut across genetic boundaries. Not only is retroflexion absent from a great many Tibeto-Burman languages of Nepal and northeast India, but also from the isolate Kusunda in Nepal, and from Assamese (Indo-Aryan), Khasi (Austro-Asiatic), and the Tai-Kadai languages of Assam, all in northeast India. Secondly, it is possible to draw a line that runs roughly east to west across Nepal, such that all Tibeto-Burman languages with retroflexion are north of the line, and all those without it are south of the line (see the dashed line in Map 3). Thus, within Nepal, Tibeto-Burman languages with retroflexion are concentrated along the northern border, whereas those without retroflexion are concentrated to the south.

In sum, the retroflex area affects all language families in the region and covers most of the Indo-Pakistani subcontinent, except for parts of Nepal and northeast India. Significantly, it extends far into China to the northeast. However, looking at retroflexion independent of manner of articulation obscures important differences between languages. The following sections examine retroflex obstruent, nasal, liquid, approximant and vowels systems independently, highlighting differences between language families and sub-regions.

4. Retroflex obstruents

The class of obstruents includes stops, affricates and fricatives. As a general rule, most South Asian languages with retroflexion have retroflex obstruents of some kind. Exceptions to this generalization include a few Iranian languages of western Pakistan (e.g., dialects of Balochi and Pashto) and some Munda languages (e.g., Sora, Gutob, Gorum, and possibly Korku). In these languages, contrast between denti-alveolar and retroflex stops is primarily limited to loanwords, while retroflex sonorants (i.e., nasals and/or liquids) are reported in native vocabulary.

Figure 4 shows the frequency of retroflex obstruents in each of the main families. Here and elsewhere, the symbols ‘t’, ‘tʂ’, and ‘ʂ’ represent natural classes of stops, affricates, and fricatives (respectively), independent of laryngeal features such as voicing and aspiration. Unaffricated retroflex stops are by far the most common type of retroflex obstruent in each family. Retroflex affricates and fricatives are rare in Indo-Aryan and Tibeto-Burman, absent altogether in Munda and other Austro-Asiatic languages, and nearly so in Dravidian. With the exception of Toda, which has a native /ʂ/, retroflex fricatives are typically limited to Sanskrit loanwords in Dravidian. The

low frequency of affricates and fricatives in all families is not something unique to retroflexion. South Asian languages have been characterized as ‘poor in fricatives’ on the whole (Ramanujan & Masica 1969; Neukom 1999). Most languages of the region have only one or two fricatives, typically /s/ and /h/, and some Dravidian languages lack them altogether.

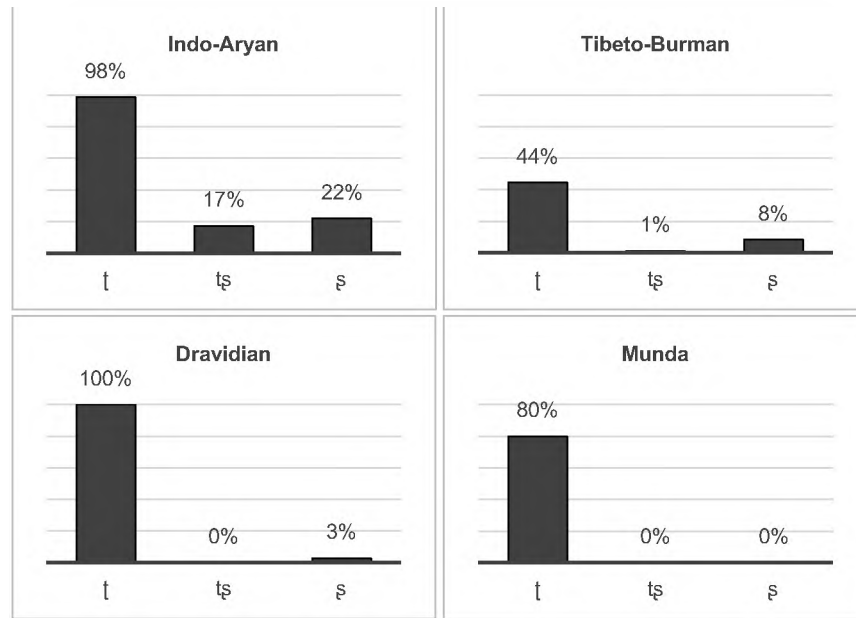


Figure 4: Percentage of languages with retroflex stops (ɖ), affricates (tʃ), and fricatives (ʃ) by family

Within the Indo-Iranian family, retroflex affricates and fricatives are most characteristic of two genetic subgroups: the Dardic branch of Indo-Aryan in northern Pakistan (86%), and the Nuristani languages in Afghanistan (100%). They also occur in Wakhi and a few other Iranian languages of the south Pamir group in Afghanistan, but not in the north Pamir languages (Tikkanen 2008). Outside of these subgroups, retroflex fricatives (and affricates) are rare in Indo-Iranian. With few exceptions, they are typically limited to Sanskrit loanwords. Notable exceptions include Dhivehi, an Indo-Aryan language of the Maldives, which has developed /ʃ/ from independent sources (Cain & Gair 2000), and the southwest dialect of Pashto (Iranian) in Afghanistan, which has /ʃ, ʒ/ as its only native retroflex obstruents (Elfenbein 1997).

Tikkanen (1999) proposes two basic retroflex inventory types, A and B, as summarized in (1). Type A has a contrast between dental and retroflex stops (with or without retroflex flaps), while Type B incorporates and augments Type A by adding a contrast between retroflex and non-retroflex (dental and laminal post-alveolar) sibilant fricatives. Each type has several possible subtypes (not shown here) based on the presence or absence of retroflex affricates, nasals, and liquids in the system. However, the basic A/B distinction is defined entirely with respect to retroflex stops and fricatives. Tikkanen (1999) demonstrates that each of these types is the result of a different evolutionary path in the development of retroflexion, and possibly a different substrate influence. Be that as it may, the typology in (1) is of limited use from a cross-linguistic perspective. This is because it excludes certain retroflex obstruent inventories that are attested, even within South Asia (e.g., a system with retroflex affricates and fricatives that does not include retroflex stops, which occurs in Tibeto-Burman). For this reason, Tikkanen's typology will not be adopted here.

(1) Tikkanen's basic retroflex typology (1999: 139, transcription modified)

Type A: $\text{ɖ} / \text{t}$ and/or $\text{ɖ} / \text{d} \pm \text{ɖ} / \text{r}$

Type B: A + $\text{ʂ} / \text{ʃ} / \text{s} \pm \text{ʐ} / \text{ʑ} / \text{z}$

With three different manners of articulation (and disregarding other features), there are seven logically possible retroflex obstruent systems: stops alone $\{\text{ɖ}\}$, affricates alone $\{\text{tʂ}\}$, fricatives alone $\{\text{ʂ}\}$, or one of four possible combinations of these: $\{\text{ɖ}, \text{tʂ}\}$, $\{\text{ɖ}, \text{ʂ}\}$, $\{\text{tʂ}, \text{ʂ}\}$, $\{\text{ɖ}, \text{tʂ}, \text{ʂ}\}$. Figures 5-6 show the relative frequency of each type among languages that have retroflex obstruents. Most of the types are attested cross-linguistically, as shown in Figure 5, but only four occur within the (narrow) South Asia sample, as shown in Figure 6. Approximately 74% of South Asian languages have retroflex obstruents of some kind. Of these, the vast majority (85%) have stops alone $\{\text{ɖ}\}$; 6% have stops and fricatives $\{\text{ɖ}, \text{ʂ}\}$; 1% have affricates and fricatives $\{\text{tʂ}, \text{ʂ}\}$; and 9% have all three $\{\text{ɖ}, \text{tʂ}, \text{ʂ}\}$. Systems with affricates or fricatives alone, $\{\text{tʂ}\}$ or $\{\text{ʂ}\}$, do not occur in any language within South Asia proper, but are reported for some languages just outside the region (see Map 4).

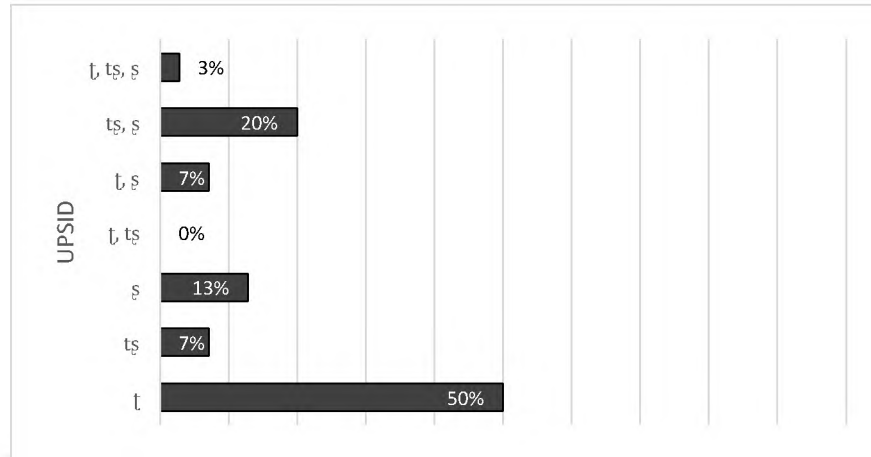


Figure 5: Frequency of retroflex obstruent system types in UPSID languages with retroflex obstruents.

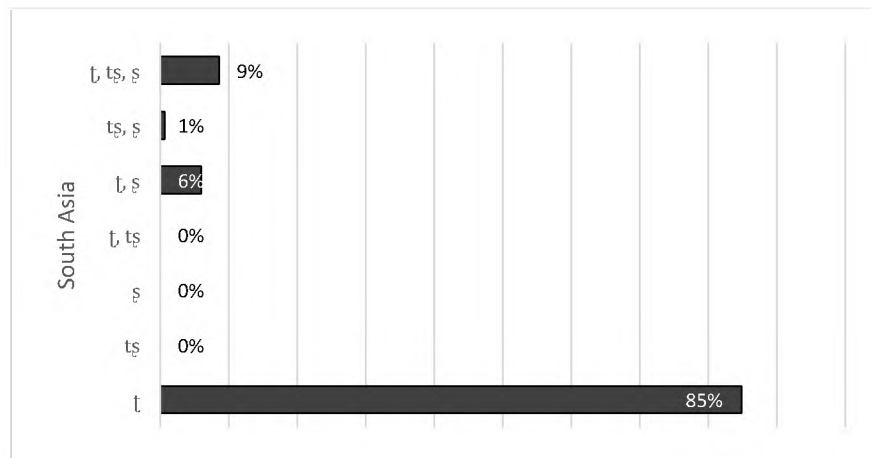


Figure 6: Frequency of retroflex obstruent system types in SA languages with retroflex obstruents.

Map 4 shows the distribution of retroflex obstruent systems in South Asia and environs. At least three areal patterns are identifiable on the map. The first covers most of the Indian subcontinent and represents the vast majority of Indo-Aryan, Dravidian, Munda and Tibeto-Burman languages that have retroflex stops as their only retroflex obstruent (cf. Tikkanen's Type A in [1] above). The second is the northwestern

periphery of South Asia, where languages tend to have retroflex fricatives and affricates *in addition to stops* (cf. Tikkanen’s Type B in [1]). Within this area, two distinct sub-patterns can be observed. One is a cluster of languages with retroflex stops and fricatives {ʈ, ʂ}, but not affricates. This cluster consists mostly of Tibeto-Burman languages of the Bodish group in northwestern India (e.g., Purik, Ladakhi, Zanskari, Spiti), but also includes Brokskat, an Indo-Aryan language of the Dardic group. The other northwestern sub-pattern is a cluster of languages with a three-way contrast between retroflex stops, affricates, and fricatives {ʈ, tʂ, ʂ}. This cluster includes most Indo-Aryan languages of the Dardic group in northern Pakistan (e.g., Dameli, Kalami, Kalasha, Khovar, Indus Kohistani, Palula, Shina, etc.), and the Nuristani languages in Afghanistan (e.g., Ashkun, Kamviri, Kati, Waigali), together with Wakhi (and other Iranian languages of the south Pamir group), and Burushaski (Isolate).

The third major area on Map 4 begins on the northeastern periphery of India and extends far into China. Like the northwestern area, languages in this area favour systems with retroflex affricates and (to a lesser degree) fricatives, but typically *without stops*. All languages in this area are Sino-Tibetan, but they do not constitute an independently established subgroup of that family. A few Tibeto-Burman languages of the Bodish and rGyalrongic groups are reported to have retroflex affricates {tʂ} as their only native retroflex obstruent (e.g., Kami, Thebo, Caodeng Rgyalrong). More often than not, languages in this area have retroflex fricatives in addition to affricates {tʂ, ʂ}. Systems of this type are reported in the Ngwi-Burmese branch of Tibeto-Burman (e.g., Lisu), the Bodish branch (e.g., varieties of Kham Tibetan, Amdo Tibetan, and Cone), the Qiangic branch (e.g., Qiang, and Niuwozi Pumi), and Mandarin Chinese. Wadu Pumi and Dayang Pumi (both Qiangic) have systems comparable to those of the Dardic/Nuristani/Burushaski type in the northwest. These systems include retroflex stops in addition to affricates and fricatives {ʈ, tʂ, ʂ}.

Diachronically, there is a correspondence between retroflex affricates in Tibeto-Burman languages to the east and retroflex stops in Tibeto-Burman languages to the west. Both evolved historically from obstruent-liquid onset clusters (Cr-). Moreover, the retroflex stops of some languages are variably pronounced with a rhotic or affricated release, as in the case of Lhasa Tibetan [ta] ~ [tʂa] ‘hair’ (< Old Tibetan *skra*), [tʰi] ~ [tʂʰi] ‘knife’ (< Old Tibetan *gri*; Beyer 1992: 28). Since affrication is not consistent or distinctive in these cases, the segments in question are typically treated as simple stops. Thus, Tibeto-Burman languages tend to have either stops {ʈ}, affricates {tʂ}, or variation between stops and affricates {ʈ ~ tʂ}, but rarely a contrast between the two (with the exception of certain varieties of Pumi mentioned above).

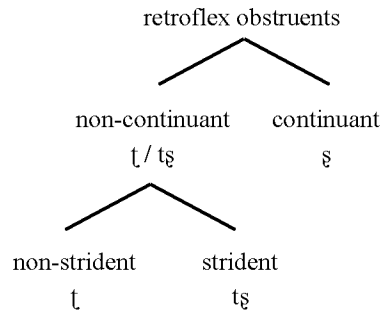
Of the seven logically possible obstruent systems identified above, only one type is completely unattested. No language in the South Asia survey (Figure 6) or the

UPSID sample (Figure 5) is reported to have retroflex stops and affricates without fricatives $\{t, tʂ\}$. Contrast between stops and affricates only occurs in systems that also have retroflex fricatives (i.e., $\{t, tʂ, ʂ\}$). This suggests the possibility of an implicational universal like that in (2). It might be the case that contrast between continuant and non-continuant retroflex obstruents ($/ʂ/$ vs. $/t/$ or $/tʂ/$) is more salient than contrast between strident (‘noisy’) and non-strident stops ($/tʂ/$ vs. $/t/$). If so, then we can represent this implicational relation by means of the feature hierarchy in (3), in which less salient contrasts apply only after more salient ones. The hierarchy predicts attested systems, including $\{t, ʂ\}$, $\{tʂ, ʂ\}$, and $\{t, tʂ, ʂ\}$, but not the unattested system $\{t, tʂ\}$.

(2) Implicational universal

If a language distinguishes retroflex stops and affricates, it also distinguishes retroflex fricatives.

(3) Feature hierarchy



In sum, contrast between retroflex and non-retroflex (dental or denti-alveolar) obstruents is limited to stops throughout most of the subcontinent. However, languages in the northern peripheries of South Asia exhibit different systems. Those in the northwest favour retroflex affricates and fricatives in addition to stops; while those in the northeast favour affricates and fricatives without stops. The rest of the paper focuses on retroflex sonorants, beginning with nasals.

5. Retroflex nasals

Most South Asian languages (about 60%) distinguish stops at five places of articulation: labial, dental, retroflex, palatal, and velar. At a phonetic level, most also have a corresponding set of nasals in homorganic nasal-stop sequences. However, languages differ with respect to the phonemic status of each nasal. Figure 8 shows the

overall frequency of the five most common nasal phonemes in South Asia. Figure 7 shows statistics for the same set in the UPSID database.

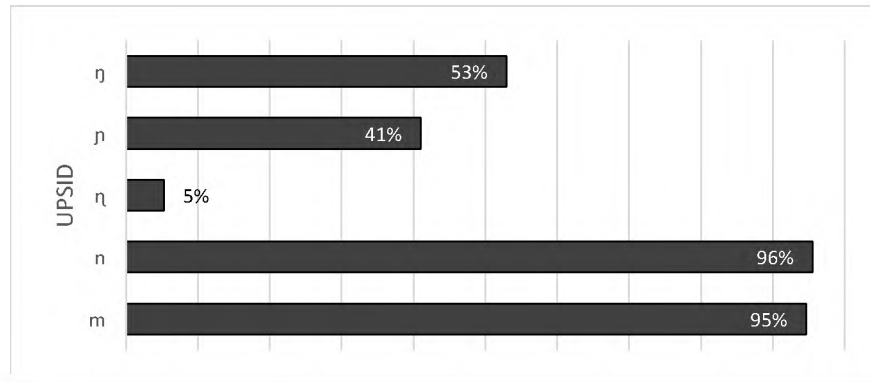


Figure 7: Percentage of languages with selected nasal phonemes in the UPSID database

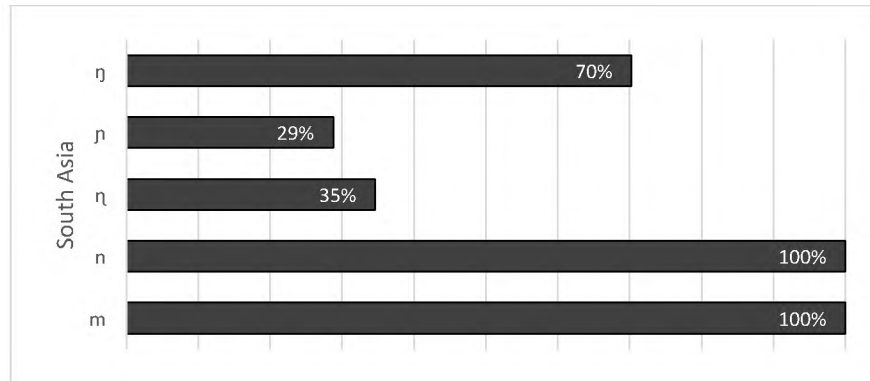


Figure 8: Percentage of languages with selected nasal phonemes in South Asia

Minimally, all South Asian languages have /m/ and /n/, but they differ with respect to retroflex /ɳ/ and other nasals. The phoneme /ɳ/ occurs in only about 5% of the world's languages, making it the least frequent of the five nasals in Figure 7. In South Asia, however, it occurs in about 35% of languages, making it more common than palatal /ɲ/ (29%), but not velar /ŋ/ (70%).

Figure 9 shows the frequency of retroflex nasals in each South Asian family. Palatal and velar nasals are included for comparison. Retroflex nasals occur as phonemes in each family, although there are significant differences between them. As

a broad generalization, we may say that Dravidian and (to a lesser extent) Indo-Aryan languages favour retroflex nasals over others, whereas Munda and Tibeto-Burman languages favour velar and palatal nasals over retroflex. In both Dravidian and Indo-Aryan, retroflex /ɳ/ is the most frequent nasal phoneme (after /m/ and /n/), occurring in more languages than either palatal /ɲ/ or velar /ŋ/. In Munda and Tibeto-Burman the situation is reversed; retroflex /ɳ/ is the least frequent nasal phoneme, occurring in fewer languages than either palatal /ɲ/ or velar /ŋ/. Within the Munda sample, /ɳ/ is reported as a phoneme exclusively in the North Munda branch (e.g., Mundari, Korwa, Kodaku, Brijia, Bhumij). Elsewhere, its phonemic status is described as marginal or doubtful, if it is listed at all. Retroflex nasals are almost non-existent in Tibeto-Burman. They are reported in only two Tibeto-Burman languages of South Asia (Kinnauri in northwest India, and Thangmi in Nepal), and in none of those surveyed outside the region. Similarly, retroflex nasals do not occur in any of the minor families and isolates.

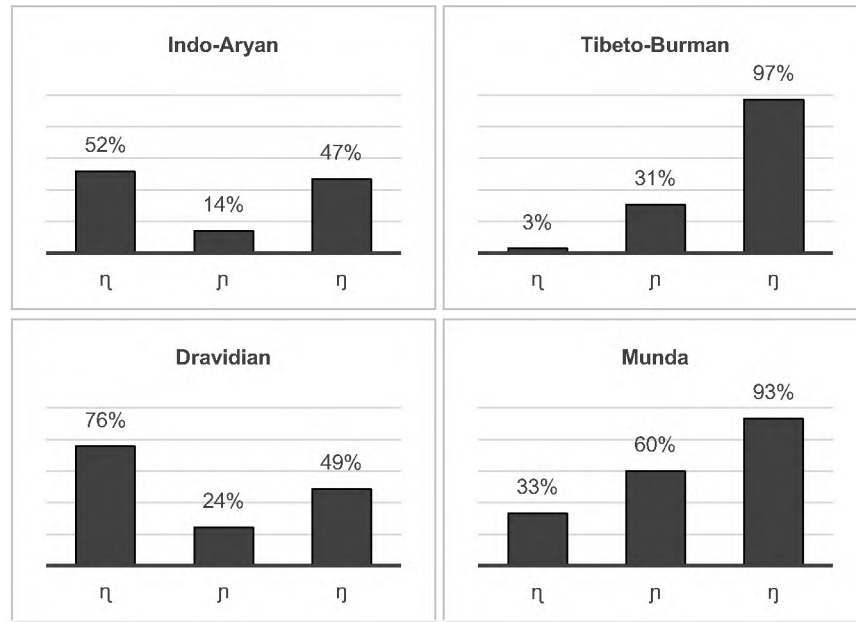


Figure 9: Frequency of retroflex /ɳ/, palatal /ɲ/, and velar /ŋ/ nasals by language family

With five nasal places of articulation, we might expect twenty-five logically possible system types (i.e., 5 x 5). However, the number of attested types is much less because all South Asian languages have /m/ and /n/. There are only eight attested types in the South Asia sample, four of which include the retroflex nasal: {m, n}, {m, n, ɳ},

{m, n, ɲ}, {m, n, ɳ}, {m, n, ɳ, ɲ}, {m, n, ɳ, ɳ}, {m, n, ɳ, ɳ}, and {m, n, ɳ, ɳ, ɳ}.⁸ The statistically dominant type in each family is listed in (4). Two percentages are listed for each family. The first represents the proportion of languages that has precisely the system listed to the left, with no additional nasals (not counting laryngeal distinctions, if they occur). The second represents the proportion that includes that system (i.e., some languages may have additional nasals).

- (4) Statistically dominant nasal place system in each family
- | | | | | | |
|------------------|---|---|---|------------|------------|
| a. Indo-Aryan | m | n | ɳ | (31 – 52%) | |
| b. Dravidian | m | n | ɳ | (35 – 76%) | |
| c. Munda | m | n | ɳ | ɳ | (47 – 60%) |
| d. Tibeto-Burman | m | n | | ɳ | (65 – 97%) |

The geographic distribution of languages with retroflex nasals is shown in Map 5, along with the distribution of velar nasals for comparison. Notice that retroflex nasals have a much narrower distribution than retroflex stops (cf. Map 4). They are concentrated in a largely uninterrupted area that stretches from the southern tip of India, through western India and Pakistan, into parts of Afghanistan in the northwest. This area represents the primary strongholds of the Dravidian and Indo-Iranian families. Retroflex nasals also appear in a slightly more diffuse cluster of languages in eastern India. This cluster includes a few eastern Indo-Aryan languages (e.g., Oriya, Desiya Oriya), some Central and South-Central Dravidian languages (e.g., Ollari Gadaba, Mudhili Gadaba, Kui, Kuvi, Pengo, Konda), and also some North Munda languages (e.g., Mundari, Korwa, Kodaku, Brijia, Bhumij).

Areas *with* retroflex nasals are not the only ones to cut across genetic lines; areas *without* them do likewise. Retroflex nasals are absent from a broad swath of languages in central India that includes Dravidian (e.g., Southern Gondi, Parji, Kolami, etc.), Indo-Iranian (e.g., Bhatia, Bahelia, Bundeli, Bhojpuri, Magahi, etc.), and Munda languages (e.g., Gta', Remo, Gutob, Korku, etc.). The area begins around southern Chhattisgarh and adjacent parts of Orissa, and extends northward through western Maharashtra, Madhya Pradesh, and Uttar Pradesh, to the Himalayas in the north. There it joins the greater area without retroflex nasals, which is rooted in Sino-Tibetan and Austro-Asiatic, and extends across the Himalayas from northwest India, through Nepal and Bhutan, to northeast India and beyond.

⁸ Additional types would be recognized if other place distinctions were considered. For example, a few South Dravidian languages distinguish alveolar nasals from both dental and retroflex (e.g., Malayalam and Paniya).

It is interesting to note that languages tend to have either retroflex or velar nasals, but rarely both. Thus, as first observed by Ramanujan and Masica (1969: 567), languages with retroflex nasals are in almost perfect complementation with those that have velar (and to a lesser degree) palatal nasals. This generalization is corroborated by the present survey. The near-complementary pattern is clearly evident in Map 5.

In sum, Dravidian and Indo-Aryan languages tend to have retroflex nasals, while Munda and Tibeto-Burman languages generally lack them. However, languages with and without retroflex nasals can be found in all families because the distribution of each type is more geographic in nature than genetic.

6. Retroflex liquids

The class of liquids encompasses trills, flaps (sometimes called taps), and laterals. Non-lateral retroflex liquids in South Asia are typically described as flaps. Retroflex trills are not common, though they are reported, for instance in Toda (Spajic et. al. 1996). However, no language is reported to distinguish retroflex trills from flaps, and even Toda speakers can vary somewhat between the two. Thus, for the purpose of this study, retroflex trills are subsumed under the category of retroflex flaps.

In environments where stops commonly undergo lenition (e.g., between vowels), retroflex stops are often pronounced as flaps. This is especially true of the voiced retroflex stop /ɖ/ and its less frequent breathy counterpart /ɖʱ/. As a result, retroflex flaps occur very frequently at a phonetic level as allophones of retroflex stops.⁹ However, secondary developments in some languages have produced minimal or near-minimal pairs for stops and flaps, so that it is often necessary to recognize the flaps as distinct phonemes from a synchronic point of view.

Like other retroflex segments, retroflex liquids are relatively marked cross-linguistically. They occur in only 10% of languages in the UPSID database. Within South Asia, however, they occur in almost half of all languages (45%). Once again there are significant differences between families and sub-regions. Figure 10 shows that retroflex liquids are most characteristic of Dravidian (89%), and least characteristic of Tibeto-Burman, where they are exceedingly rare (3%). Only two Tibeto-Burman languages in the survey are reported to have retroflex liquids: Purik and Spiti in northwest India. Retroflex liquids of one kind or another also occur in the majority of Indo-Aryan (63%) and Munda languages (80%). Among the ‘other’ minor families and isolates, they occur only in some Andamanese languages (e.g., Jarawa and Onge).

⁹ In fact, retroflex nasals and laterals are also commonly flapped under conditions of lenition. However, no language maintains a contrast between flapped and non-flapped retroflex nasals or laterals.

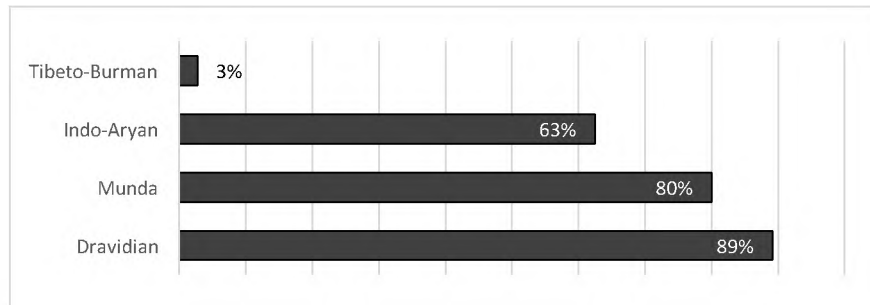


Figure 10: Percentage of languages with retroflex liquids in each South Asian family

Taking retroflex flaps and laterals as our basic liquid types, there are three possible systems: flaps alone {ɽ}, laterals alone {ɭ}, or both {ɽ, ɭ}. Each of these types is attested in South Asia. Figure 11 shows the frequency of each type in each of the main families. Retroflex laterals do not occur in Tibeto-Burman. Otherwise, both flaps and laterals are attested to some degree in each family. Dravidian favours retroflex laterals over flaps, whereas Indo-Aryan and Munda favour retroflex flaps over laterals. Within Munda, only Juang is reported to have a retroflex lateral. Simpler systems containing a flap or lateral are far more common than complex systems containing both, which are reported only for a handful of Dravidian and Indo-Aryan languages.

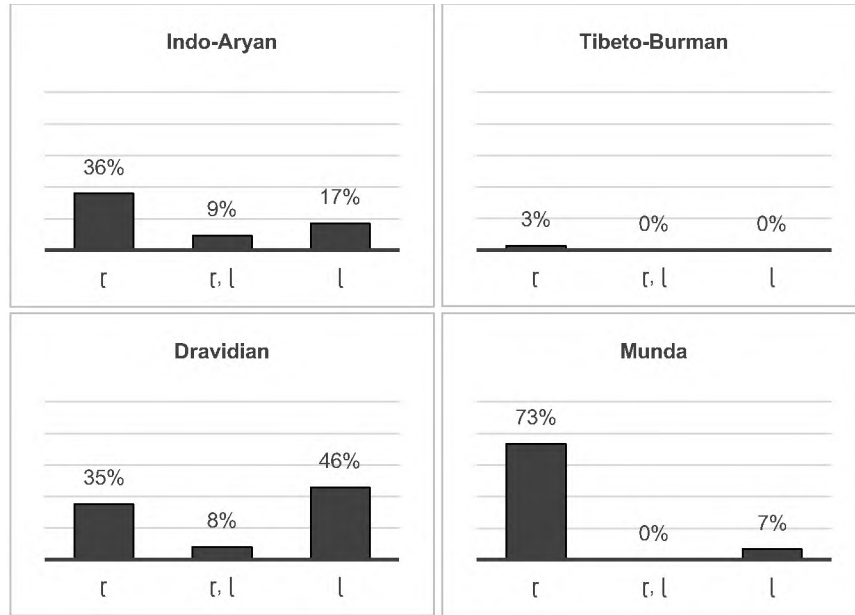


Figure 11: Percentage of languages with retroflex flaps (ɽ), laterals (l), or both, by family.

All South Asian languages have a dental/alveolar lateral /l/, and 96% have some form of dental/alveolar /r/, typically described as a flap or trill.¹⁰ Adding these to the equation, we can compute the statistically dominant liquid system for each family, as shown in (5). Once again, two different percentages are listed per family. The first represents the proportion of languages that has precisely the system listed to the left, with no additional liquids (not counting laryngeal distinctions, if they occur). The second represents the proportion that includes that system (i.e., languages that have additional liquids are also counted).

- (5) Statistically dominant liquid system in each family
- | | | | | |
|------------------|---|---|---|------------|
| a. Indo-Aryan | r | l | ɽ | (34 – 45%) |
| b. Dravidian | r | l | | (30 – 54%) |
| c. Munda | r | l | ɽ | (73 – 73%) |
| d. Tibeto-Burman | r | l | | (86 – 89%) |

The distribution of the various retroflex liquid systems is shown in Map 6. Here again, each type cuts across genetic boundaries, demarcating its own geographic region.

¹⁰ Languages without /r/ are all Tibeto-Burman or Tai-Kadai languages of northeast India.

Systems with a retroflex lateral {ʎ} dominate the south in an area that begins with Dhivehi (Indo-Aryan) in the Maldives, and Tamil (Dravidian) in northern Sri Lanka and southern India, and extends northward through western India as far as Kangri (Indo-Aryan) in Himachal Pradesh. With few exceptions, just about every Dravidian and Indo-Aryan language along this path has retroflex laterals. This area closely resembles that of retroflex nasals (cf. Map 5), except that the nasals extend much farther westward into Pakistan and Afghanistan, whereas the laterals stop somewhere around the India-Pakistan border.

While retroflex laterals dominate the south, systems with retroflex flaps {ɽ} dominate the north. They cover a large area that includes most of Pakistan and eastern Afghanistan, and extends (south)eastward across north and central India, all the way to the Bay of Bengal. This broad swath harbours representatives of all the main families, including Iranian (e.g., Balochi, Pashto, etc.), Nuristani (e.g., Ashkun, Waigali, etc.), Indo-Aryan (e.g., Sindhi, Panjabi, Hindi, Bhojpuri, Bengali, and a host of others), North Dravidian (e.g., Brahui, Kurux, Malto), Central Dravidian (e.g., Ollari Gadaba, Parji), all of South-Central Dravidian, except for Telugu (e.g., Gondi, Konda, Kui, Kuvi, etc.), and most Munda languages (e.g., Mundari, Ho, Santali, Gta', Sora, etc.). Its northern frontier even includes two Tibeto-Burman languages, Purik and Spiti, which are the only representatives of that family with retroflex liquids.

The retroflex lateral and flap areas overlap in western India. Thus, there is a cluster of Indo-Aryan languages centred in and around the states of Rajasthan and Punjab that distinguishes retroflex flaps and laterals {ɽ, ʎ}. Examples include Marwari, Shekawati, Bagri, Panjabi, Haryanvi, and Kangri. There is also a small cluster of South Dravidian languages in the Nilgiri Hills region with essentially the same system. This group includes Toda, Kota, and Irula.

In sum, retroflex liquids occur in most Indo-Aryan, Dravidian, and Munda languages, but they are not characteristic of Tibeto-Burman. Indo-Aryan and Munda show a preference for /ɽ/ over /ʎ/, while Dravidian shows a preference for /ʎ/ over /ɽ/. However, each of these systems constitutes an isogloss that cuts across genetic lines. Retroflex laterals dominate southern and western India, while flaps dominate the north. The two areas overlap in western India, where some languages distinguish both flaps and laterals.

7. Retroflex approximants and vowels

Retroflex approximants are transcribed variously as [ɻ] or [ɭ]. Strictly speaking, the first symbol represents a retroflex approximant while the second represents an apical/alveolar approximant. However, even segments transcribed with [ɻ] are often

described as having a retroflex quality, and (to the best of my knowledge) no language distinguishes the two. Thus, for the present study both are regarded as retroflex.

Retroflex approximants and vowels are exceedingly rare at a phonological level. Only 3.5% of languages in UPSID distinguish retroflex /ɭ/ from some type of non-retroflex rhotic /r/ (almost all of them Australian), and less than 1% distinguish retroflex vowels.¹¹ The picture is not much different in South Asia, where the corresponding figures are just 2.4% for /ɭ/ (or /ɻ/) and 1% for retroflex vowels.

Proto-Dravidian had a retroflex approximant *ɭ (often transcribed *ʒ). It is preserved only in regional and social dialects of Tamil and Malayalam. Elsewhere, it has merged with one of the liquids or /j/, or dropped out altogether (Krishnamurti 2003). Within Indo-Iranian, retroflex approximants occur primarily in the Nuristani branch, where they occur either as independent phonemes (e.g., Kati, Kamviri) or as allophones of /r/ or /ɻ/ (e.g., Ashkun). Some Nuristani languages also have a nasalized retroflex approximant that contrasts with its oral counterpart and/or with /ŋ/ (e.g., Kamviri, Waigali). The phoneme /ɭ/ is also reported in two other nearby languages: Dameli, an Indo-Aryan language of the Dardic group, and the isolate Burushaski.

Contrast between /r/, on the one hand, and /ɭ/ or /ɻ/ on the other, is not reported for any Austro-Asiatic or Tibeto-Burman language in the present survey.¹² However, some Tibeto-Burman and Tai-Kadai languages of northeast India, Myanmar, and China have an apical/retroflex approximant, [ɹ] or [ɻ], as the primary phonetic realization of the phoneme /r/. Some languages also have a voiceless counterpart realized as [ɹ̥] (or even [ɻ̥]). Approximant realizations of /r/ are reported in Ao, Angami, Chokri, Rawang, Wadu Pumi, Mandarin, and many other languages. However, in such cases the retroflexion is not phonologically significant because there is no opposition between the segment in question and some non-retroflex dental or alveolar counterpart.

At a purely phonetic level, retroflex vowels occur regularly in languages that have retroflex consonants, especially in VC transitions as a result of coarticulation with a following retroflex segment. This is so pervasive in South Asia as to hardly warrant mention in most descriptions. However, contrast between retroflex and non-retroflex vowels is only reported for two languages in the South Asia sample. The first is Badaga,

¹¹ I have not counted Chamorro (West Malayo-Polynesian) as having /ɭ/, though it is listed as such in UPSID, because the language does not distinguish /ɭ/ from another non-retroflex rhotic.

¹² Contrast between an alveolar flap /r/ and a retroflex approximant /ɻ/ (or /ɭ/) may occur in Puroik (aka. Sulung), a Tibeto-Burman language of Arunachal Pradesh, India. Two speakers I recorded in 2013 consistently distinguished these sounds in minimal pairs (e.g., [araŋ] ‘sickness’, [aɻaŋ] ‘inside, under, below’). The language also has a retroflex vowel [ɔ̣], possibly a syllabic variant of /ɻ/ (e.g., [ɔ̣pua] ‘boundary sign’). Puroik is not included in the present survey because I did not have access to a complete and reliable phonological description as of the time of writing.

a South Dravidian language of the Nilgiri area. Emeneau (1939) recorded a three-way distinction for Badaga vowels: non-retroflexed, half-retroflexed, and fully retroflexed. The retroflex vowels evolved under the influence of a following retroflex liquid. Subsequent loss of the conditioning liquid promoted the retroflex vowels to phonemic status. However, it is not clear that the contrast has survived. More recent work on the language found no trace of the retroflex vowels in the speech of younger informants (Hockings and Pilot-Raichoor 1992). It is worth noting that many other Dravidian languages of Nilgiri area have developed ‘centralized’, ‘retracted’ or ‘back unrounded’ vowels under similar conditions (Diffloth 1975: 55), but it is not clear whether any of these might qualify as retroflex.

The only other case of retroflex vowel phonemes in South Asia is Kalasha, an Indo-Aryan language of the Dardic group in northern Pakistan. In Kalasha, every oral and nasal vowel has a phonemic retroflex counterpart, as shown in (6). Like the retroflex vowels of Badaga, those of Kalasha evolved under the influence of a neighboring retroflex consonant that was subsequently lost. In fact, Morgenstierne (1973) appears to have recorded a retroflex approximant (/ʁ/ in his transcription) in environments where subsequent studies have recorded retroflex vowels (Heegård and Mørch 2004). The symmetry and complexity of the Kalasha system make it typologically interesting. Cross-linguistically, retroflex vowel phonemes typically form a reduced subset of the non-retroflex system. In many cases, this subset is limited to a single retroflex (or ‘rhotacized’) vowel (often transcribed /ə̃/). The Kalasha system, with its fully symmetrical set of retroflex vowels, may be unparalleled.

(6) Kalasha vowel phonemes (Heegård and Mørch 2004)

a. non-retroflex (oral and nasal)

i	u	ĩ	ũ
e	o	ẽ	õ
a		ã	

b. retroflex (oral and nasal)

ị	ụ	ị̃	ụ̃
ẹ	ọ	ẹ̃	ọ̃
ạ		ạ̃	

Map 7 shows the geographic distribution of languages with retroflex approximants and vowels. Unlike the preceding maps, this one includes both phonetic and phonological data. On the legend, the symbols ‘ɹ/ɻ’ represent the class of apical and/or retroflex approximants as a whole, irrespective of phonemic status, voicing,

nasalization, etc. Similarly, the symbols ‘ə/ɭ’ represent the class of retroflex vowels as a whole, regardless of phonemic status, vowel quality, friction, etc.

Map 7 demonstrates that languages with retroflex approximants and vowels are concentrated in three areas corresponding to the three extremities of the South Asian ‘triangle’. Each area is associated with a different genetic group. The first area is the southern tip of India where we find Dravidian languages, such as Tamil and Malayalam, with /ɭ/, along with the retroflex vowels of Badaga in the Nilgiri hills.

The second area is the northwest periphery of South Asia associated primarily with Indo-Iranian. Here we find retroflex approximants in Nuristani languages, Indo-Aryan (Dardic) Dameli, and the isolate Burushaski. This region is also home to Kalasha with its remarkable retroflex vowel system. These phenomena are interrelated. It was the development and subsequent loss of /ɭ/ that ultimately produced the retroflex vowels of Kalasha (Heegård and Mørch 2004). Not surprisingly, then, we find that retroflex vowels are quite prominent *phonetically* in languages of the region that have /ɭ/. Thus, Perder (2013) notes that the retroflex approximant of Dameli is primarily realised as retroflexion on adjacent vowels, and that it “might be possible to analyse each vowel affected by this sound as a separate phoneme, in effect a series of retroflex vowels, as has been done for neighbouring Kalasha” (30). Similarly, with respect to the Nuristani language, Waigali, Strand (2011) reports that the phonemes /ɭ/ and /ĩ/ (/ɾ, ñ/ in his transcription) are realized “as mere ‘retroflexion’ of a following non-front vowel, without any consonantal turbulence”. Thus, there is a close affinity between the occurrence of retroflex approximants and vowels in the northwest region.

The third area of interest is the northeastern periphery of South Asia and adjacent parts of China and Myanmar. This area is anchored in Tibeto-Burman. As mentioned above, retroflexion is not phonologically significant for approximants in Tibeto-Burman. Nevertheless, languages of this area often realize /ɾ/ as an apical/retroflex approximant *phonetically*. All such cases that were reported in the literature survey are shaded accordingly on the map. This region is also home to Qiang and Mandarin (beyond the borders of the map), both of which have retroflex vowels that are lexically contrastive. Other languages of the area have a so-called retroflex ‘fricative vowel’, often transcribed [ɭ], which is effectively syllabic [z]. These fricative vowels are often treated as allophones of a high central vowel /i/ after retroflex affricates. Languages of this type include Lisu and varieties of Pumi, some of which also realize /ɾ/ as [ɭ] (e.g., Wadu Pumi, Niuwozi Pumi).

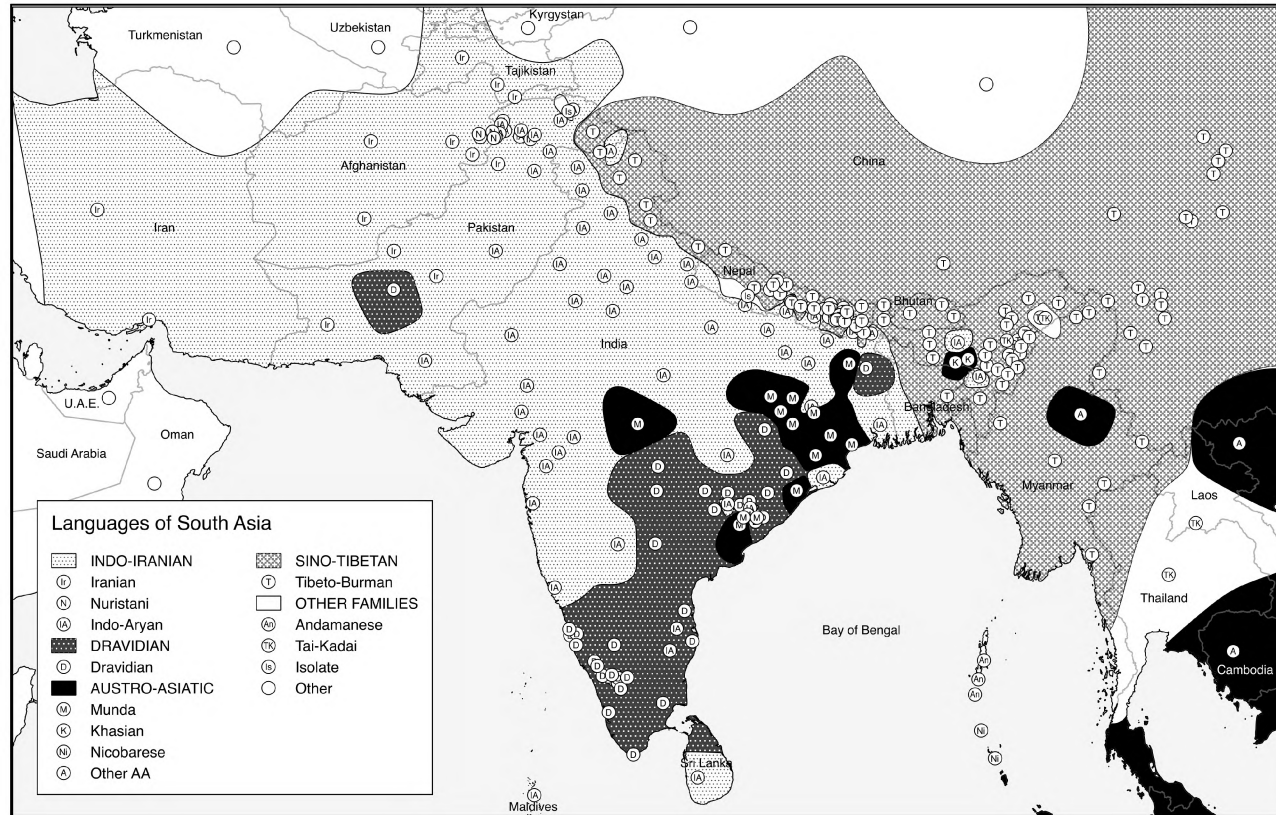
While some languages have both retroflex approximants and vowels at a phonetic level, and other languages have one or the other as a phoneme, no language has both as phonemes. This is not surprising given the diachronic relation between them: it is precisely the loss of a retroflex approximant (or liquid) that typically gives rise to

retroflex vowel phonemes. Moreover, from a synchronic point of view, approximants and vowels are often different analytical interpretations of the the same phonetic reality. Just as [j] and [w] can often be analyzed as [i] and [u], and *vice versa*, so the retroflex segments in question can be analyzed as approximants or vowels. Rarely (if ever) would the analyst need to posit both as phonemes.

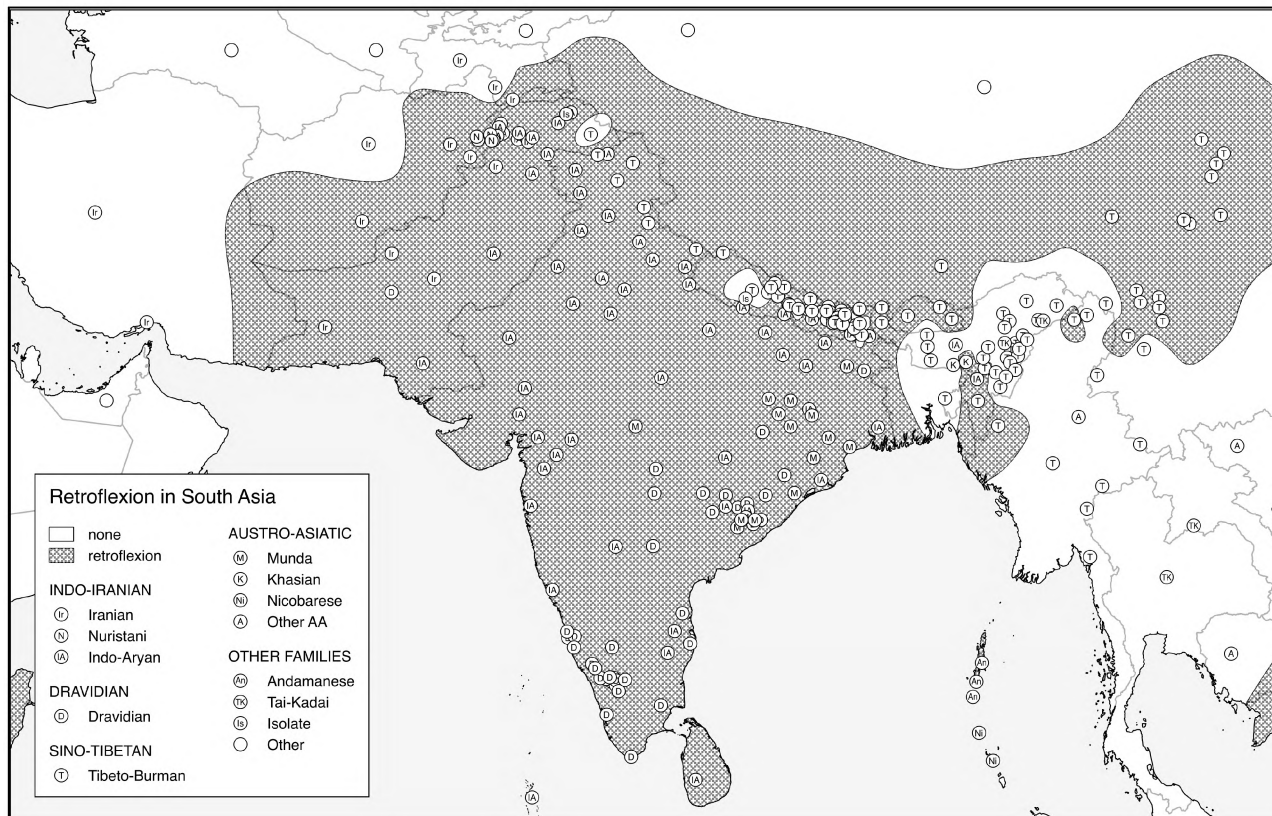
8. Conclusion

The present survey of retroflexion in South Asia corroborates many of the key observations made by Ramanujan and Masica (1969) and Tikkanen (1999), refining them wherever possible. The most general of these observations is that retroflexion is an areal feature that cuts across genetic boundaries and marks out a geographic area corresponding very closely to South Asia, with the exception of parts of Nepal and Northeast India. However, the study also highlights the fact that retroflexion extends far beyond the limits of South Asia into East Asia (China), a point that is not clearly acknowledged in the earlier studies. Tikkanen (1999) observed that retroflex systems in languages on the northwest periphery of South Asia are typologically distinct from those in the greater part of the subcontinent because they favour retroflex fricatives and affricates in addition to stops. To this we can add that languages on the northeast periphery of the retroflex area, while similar to those in the northwest, are also distinct; they favour retroflex affricates and fricatives without stops.

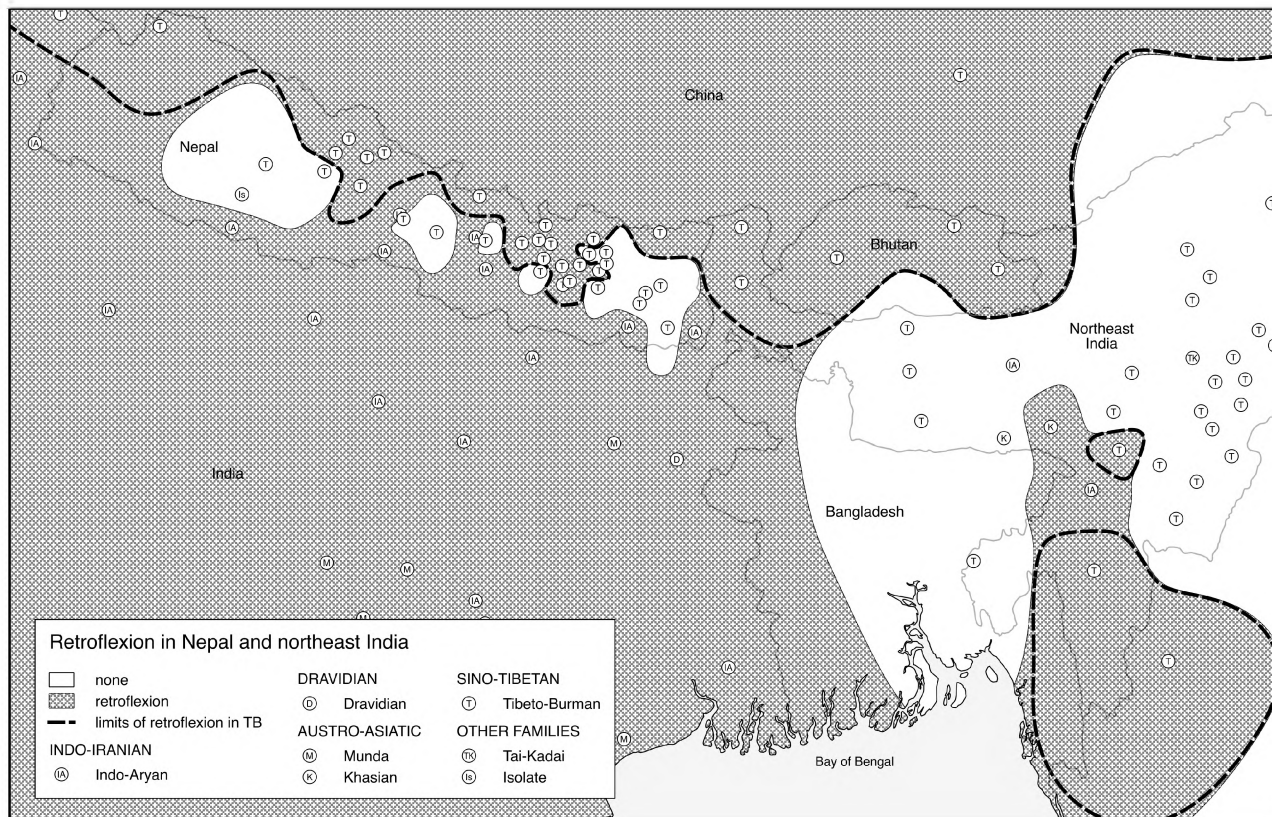
Perhaps the most striking fact about the various retroflex subsystems is that each one tends to cut across genetic lines marking out its own geographic space. It is possible to establish a few broad statistical correlations between specific types of contrast and individual language families (e.g., Dravidian favours /ɭ/ over /tʃ/; Tibeto-Burman generally lacks retroflex nasals and liquids; etc.). However, the distribution of most subsystems is clearly more geographic than genetic in nature. It remains to be seen whether these geographic areas can be correlated with other factors, linguistic or otherwise, that might shed more light on their origins.



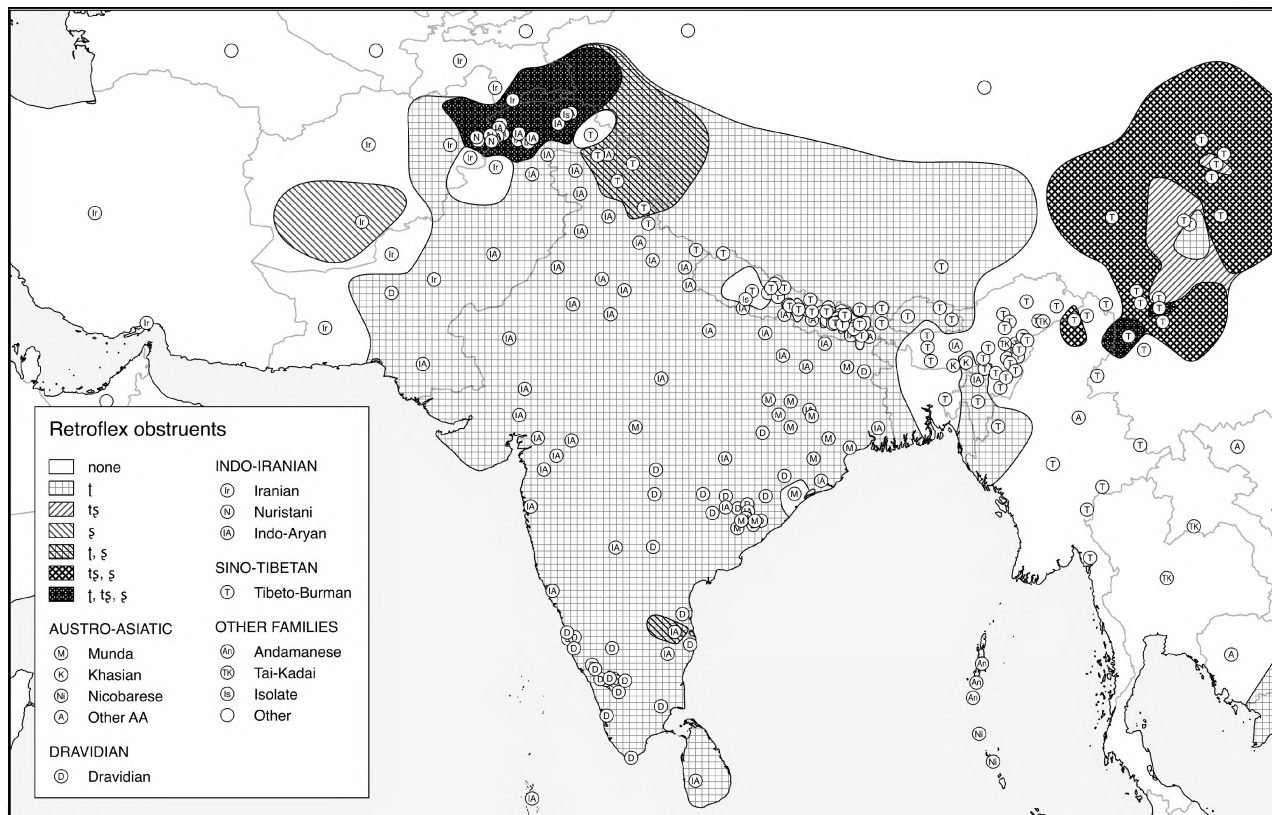
Map 1: Languages of South Asia



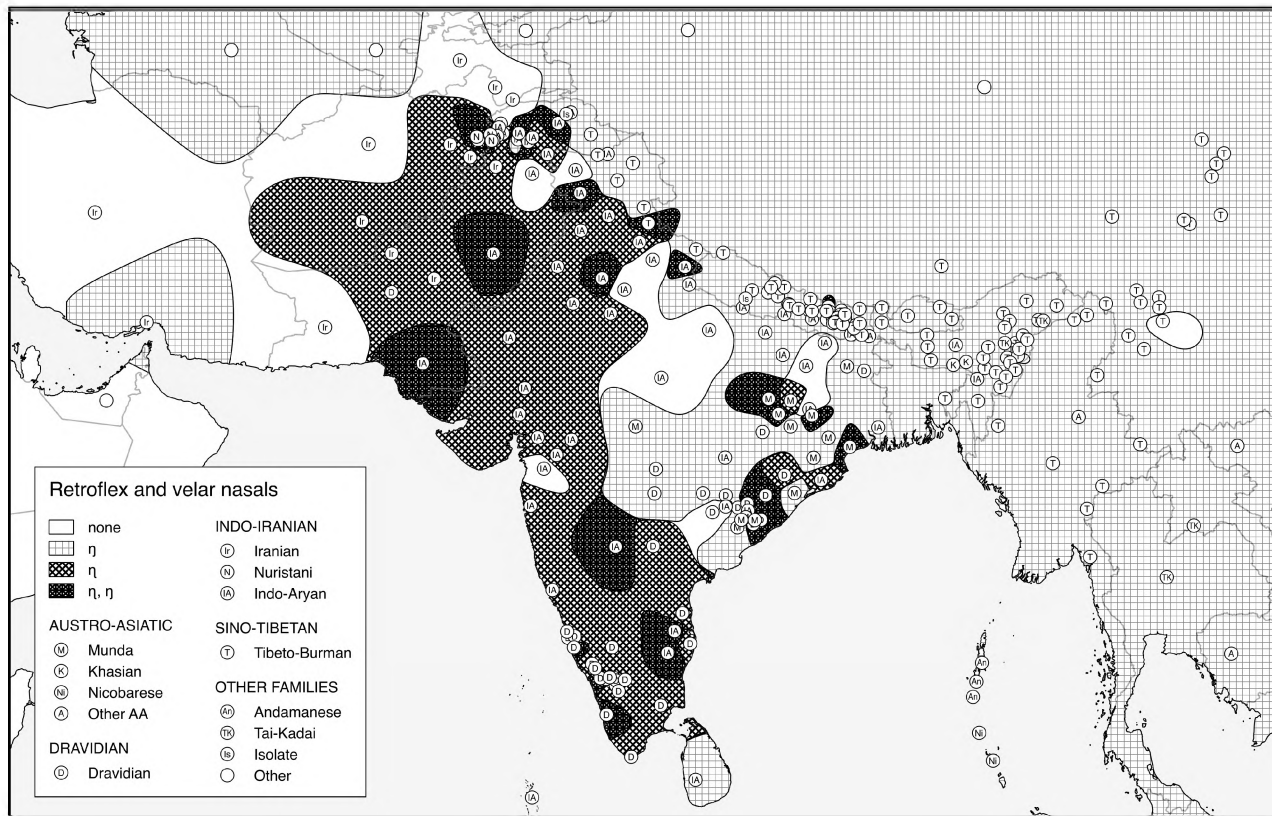
Map 2: Distribution of languages with and without retroflexion



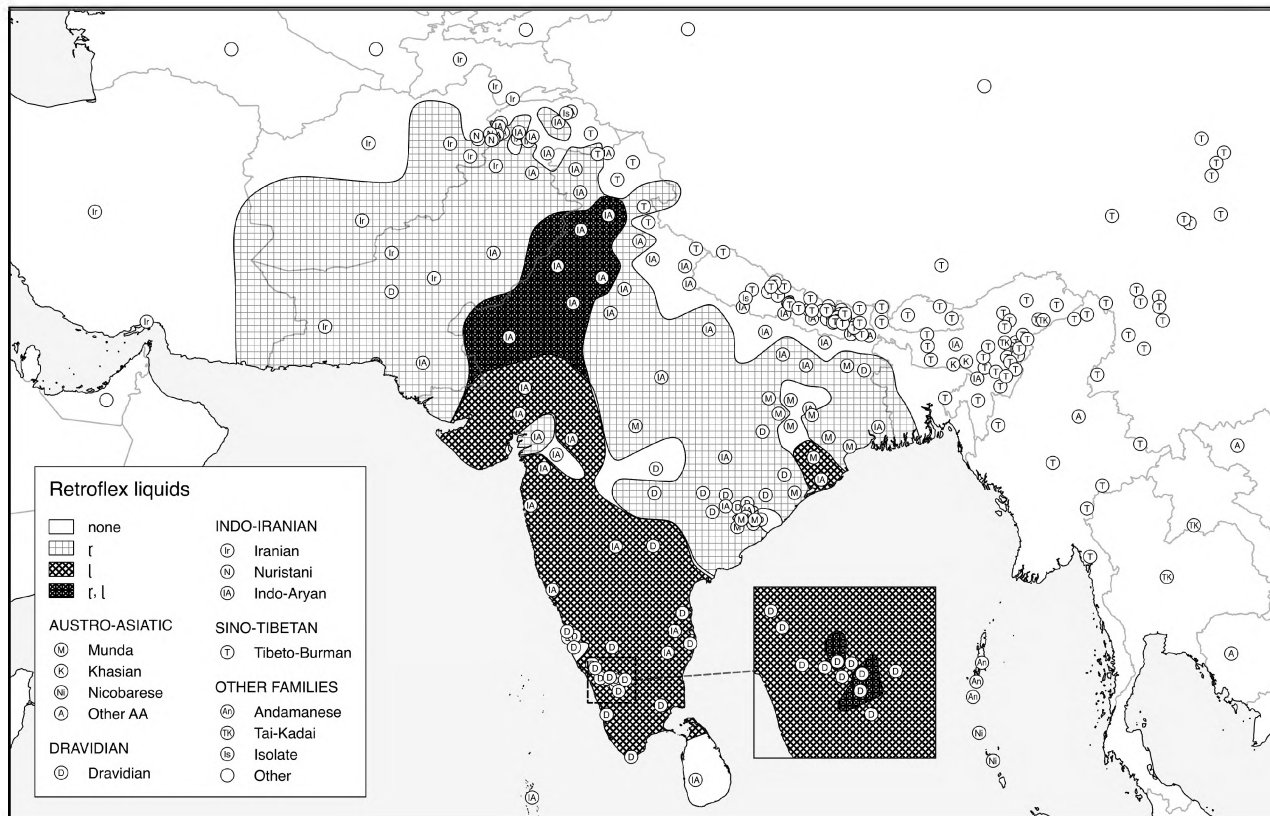
Map 3: Retroflexion in Nepal and northeast India (with dashed line showing the limits of retroflexion in Tibeto-Burman)



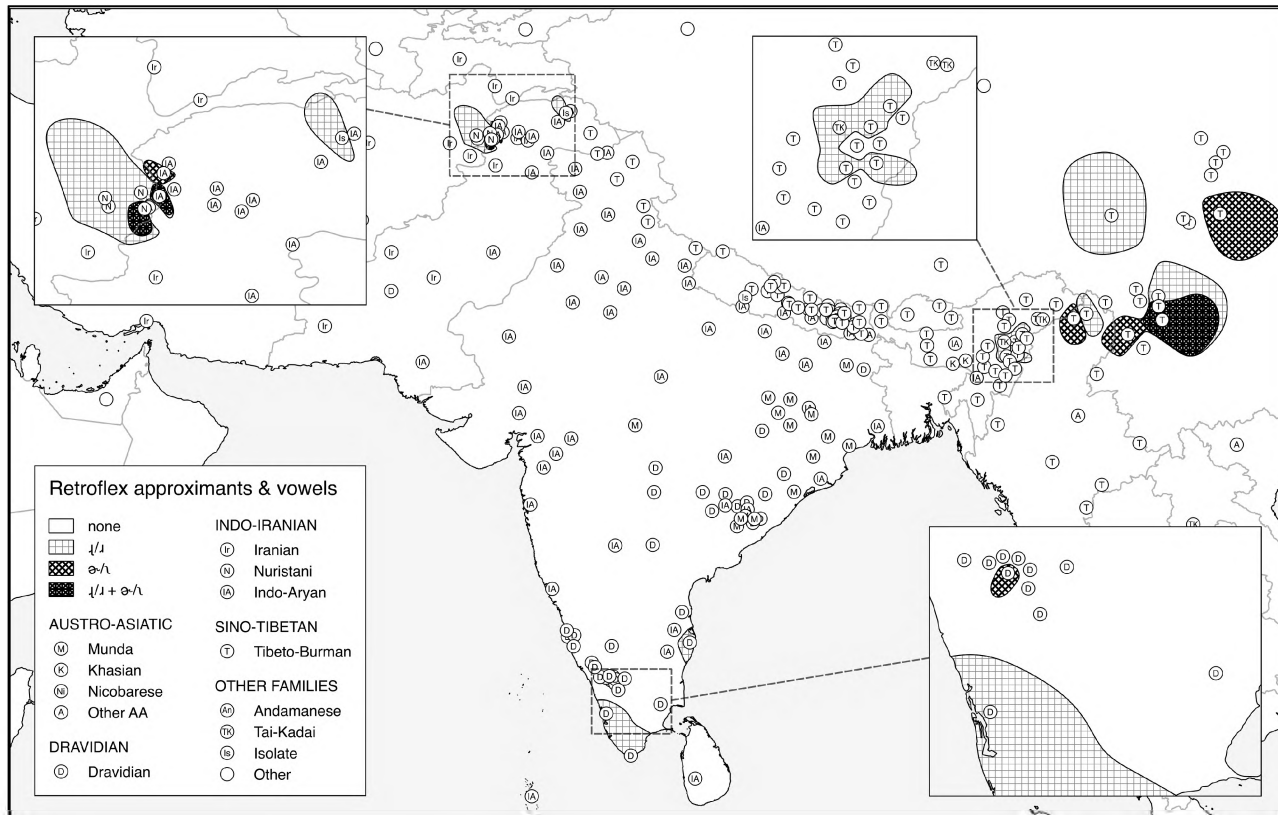
Map 4: Distribution of retroflex obstruents (stops, affricates, and fricatives)



Map 5: Distribution of retroflex and velar nasals



Map 6: Distribution of retroflex liquids (laterals and flaps)



Map 7: Distribution of retroflex approximants and vowels.

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Appendices

This appendix provides information on the languages and data sources consulted for the typological survey in the paper *Retroflexion in South Asia: Typological, genetic, and areal patterns*. The appendix consists of two lists: one covering the languages surveyed within South Asia, and the other covering those surveyed in the regions surrounding South Asia. Each list is organized alphabetically by language name. The *Ethnologue*'s three-letter ISO code is also provided for each language (as best as I can determine it), along with the language's genetic classification and data source(s). Some language names may correspond to more than one ISO code or *vice versa*. The following abbreviations are used for language classification:

AA	Austro-Asiatic	Iir	Indo-Iranian	Nu	Nuristani
An	Andamanese	Ir	Iranian	Pa	Palaungic
Ch	Chinese	Kh	Khasian	ST	Sino-Tibetan
Dr	Dravidian	Mu	Munda	TB	Tibeto-Burman
IA	Indo-Aryan	Ni	Nicobarese	TK	Tai-Kadai

In many cases, multiple sources were consulted for a given language variety. Only the primary source (or sources) are listed here. Full bibliographic details for the data sources are provided in the list of references at the end of the appendix.

Appendix A: South Asia language sample

This section lists 205 language varieties that are spoken largely (if not exclusively) within one or more of the South Asian countries (i.e., India, Pakistan, Nepal, Bhutan, Bangladesh, Sri Lanka, and Maldives). Statistics concerning South Asian languages are based on data from these languages and sources.

Language	ISO	Class	Sources
Aiton	aio	TK	Morey 2005
Andamanese, Great	abj, etc.	An	Manoharan 1989; Abbi 2006
Angami, Khonoma	njm	ST, TB	Blankenship et. al. 1993
Ao Naga, Chungli	njo	ST, TB	Gowda 1972
Ao Naga, Mongsen	njo	ST, TB	Coupe 2007
Apatani	apt	ST, TB	Abraham 1985
Assamese	asm	Iir, IA	Goswami & Tamuli 2003
Athpare	aph	ST, TB	Ebert 1997
Awadhi	awa	Iir, IA	Saksena 1937
Badaga	bfq	Dr	Hockings & Pilot-Raichoor 1992

Language	ISO	Class	Sources
Bagri	bgq	Iir, IA	Gusain 2000
Balochi, Eastern	bgp	Iir, Ir	Elfenbein 1997a; Jahani & Korn 2009
Balochi, Western/Southern	bgn, bcc	Iir, Ir	Elfenbein 1997a; Jahani & Korn 2009
Balti	bft	ST, TB	Backstrom 1994
Bareli, Pawri	bfb	Iir, IA	Immanuel & Jane 2003
Bareli, Rathwi	bgd	Iir, IA	Varkey & Vinod 2003
Belhare	byw	ST, TB	Bickel 2003
Bengali	ben	Iir, IA	Bhattacharya 1988; Dasgupta 2003
Bhatri	bgw	Iir, IA	Kirivasan & Amirthamary 2000
Bhil, Dungra	duh	Iir, IA	Mathew & Susan 2000
Bhojpuri, Northern	bho	Iir, IA	Shukla 1981
Bhojpuri, Southern	bho	Iir, IA	Verma, M 2003
Bhumij	unr	AA, Mu	Ramaswami 1992
Bishnupriya	bpy	Iir, IA	Sinha 1981
Bodo (Boro)	brx	ST, TB	Bhattacharya 1977
Brahui	brh	Dr	Elfenbein 1997b
Brijia (Asuri)	asr	AA, Mu	Sahu 1980
Brokskat	bkk	Iir, IA	Ramaswami 1975; Bashir 2003
Bundeli	bns	Iir, IA	Jaiswal 1962
Burushaski	bsk	Isolate	Anderson 1997
Byangsi	bee	ST, TB	Trivedi 1991
Camling	rab	ST, TB	Ebert 2003
Chantyal	chx	ST, TB	Noonan 2003a
Chepang	cdm	ST, TB	Caughley 1970
Chokri	nri	ST, TB	Bielenberg & Nienu 2001
Dameli	dml	Iir, IA	Perder 2013
Darai	dry	Iir, IA	Kotapish & Kotapish 1973
Deori	der	ST, TB	Jacquesson 2005
Desiya Oriya	dso	Iir, IA	Mathews 2003
Dhanki	dhn	Iir, IA	Kulkarni 1976
Dhanwar	dhw	Iir, IA	Kuegler & Kuegler 1974
Dhimal	dhi	ST, TB	Cooper 1999
Dimasa	dis	ST, TB	Misra 1986
Divehi (Maldivian)	div	Iir, IA	Cain & Gair 2000
Dogri	dgo	Iir, IA	Ghai 1991
Domaaki	dmk	Iir, IA	Lorimer 1939
Dumi	dus	ST, TB	van Driem 1993
Dzongkha	dzo	ST, TB	van Driem 1992

Language	ISO	Class	Sources
Gadaba, Mudhili	gau	Dr	Bhaskararao 1980, 1998
Gadaba, Ollari	gdb	Dr	Bhattacharya 1957
Gallong (Galo Adi)	adl	ST, TB	Gupta 1963
Garasia	gra	Iir, IA	Patel 1999
Garhwali	gbm	Iir, IA	Chandrasekhar 1970
Garó (Mande)	grt	ST, TB	Burling 2003
Gondi, Muria	emu, mut	Dr	Steever 1998a
Gondi, Southern	ggo	Dr	Subrahmanyam 1968; Lincoln 1969
Gta' (Gta?)	gaq	AA, Mu	Anderson 2008
Gujarati	guj	Iir, IA	Mistry 1997
Gurung	gvr, ggn	ST, TB	Glover 1970
Gutob (Gadaba)	gbj	AA, Mu	Griffiths 2008
Hakha Lai	cnh	ST, TB	Peterson 2003
Haryanvi	bge	Iir, IA	Singh 1970
Hayu (Wayu)	vay	ST, TB	Michailovsky 2003
Hindi	hin	Iir, IA	Ohala 1994; Kaye 1997
Hmar	hmr	ST, TB	Baruah & Bapui 1996
Ho	hoc	AA, Mu	Anderson, Osada & Harrison 2008
Humla Bhotia, Limi	hut	ST, TB	Wilde 2001
Irula	iru	Dr	Periyalwar 1979; Zvelebil 1973
Jarawa	anq	An	Abbi 2006
Jaunsari	jns	Iir, IA	Satish 1990
Jero (Jerung)	jee	ST, TB	Opgenort 2005
Jirel	jul	ST, TB	Strahm & Maibaum 1971
Juang	jun	AA, Mu	Patnaik 2008
Kagate	syw	ST, TB	Hoehlig & Hari 1976
Kalami (Kalam Kohistani)	gwc	Iir, IA	Baart 1997
Kalasha	klk	Iir, IA	Heegard & Morch 2004
Kangri	xnr	Iir, IA	Sharma 1974
Kannada	kan	Dr	Sridhar 1990
Karbi (Mikir)	mjlw	ST, TB	Jeyapaul 1987
Kasaba (Irula)	iru	Dr	Pillai 1976
Kashmiri	kas	Iir, IA	Wali & Koul 1997
Khaling	klr	ST, TB	Toba & Toba 1972
Kham	kjl	ST, TB	Watters 2002
Khamyang	ksu	TK	Morey 2005
Kharía	klr	AA, Mu	Peterson 2011
Khasi	kha	AA, Kh	Nagaraja 1989

Language	ISO	Class	Sources
Khowar	khw	Iir, IA	Bashir 2003
Kinnauri	kfk	ST, TB	Sharma 1988
Kodaku	ksz	AA, Mu	Kuriakkose & Liju 2008
Kodava	kfa	Dr	Balakrishnan 1976; Ebert 1996
Kohistani, Indus	mvv	Iir, IA	Hallberg & Hallberg 1999
Koi	kkt	ST, TB	Lahaussais 2009
Kok Borok (Tripuri)	trp	ST, TB	Karapurkar 1972
Kolami, Northwestern	kfb	Dr	Emeneau 1961; Subrahmanyam 1998
Konda (Kubi)	kfc	Dr	Krishnamurti & Benham 1998
Konkani	knn	Iir, IA	Miranda 2003
Koraga, Mudu	vmd	Dr	Bhat 1971
Koraga, Onti	kfd	Dr	Bhat 1971
Koraga, Tappu	kfd	Dr	Bhat 1971
Korku	kfq	AA, Mu	Zide 1960, 2008
Korwa	kfp	AA, Mu	George & Joseph 2008
Kota	kfe	Dr	Subbaiah 1986
Kui	kxu	Dr	Winfield 1928
Kulung	kle	ST, TB	Tolsma 2006
Kumauni	kfy	Iir, IA	Apte & Pattanayak 1967; van Riesen 2000
Kundal Shahi	shd	Iir, IA	Rehman & Baart 2005
Kurtöp	xkz	ST, TB	Hyslop 2008
Kurumba Kannada (Coimbatore)	kfi	Dr	Ernest & Ernest 2000
Kurumba Kannada (Pudukkottai)	kfi	Dr	Varma 1978a
Kurumba, Betta	xub	Dr	Selvaraj & Selvaraj 2003
Kurux	kru	Dr	Pfeiffer 1972
Kusunda	kgg	Isolate	Watters 2006
Kuvi	kxv	Dr	Reddy et. al. 1974; Israel 1979
Kyerung	kgv	ST, TB	Huber 2005
Ladakhi	lbj	ST, TB	Koshal 1976
Lamani	lmn	Iir, IA	Trail 1970
Lepcha	lep	ST, TB	Plaisier 2007
Lhomi	lhm	ST, TB	Vesalainen & Vesalainen 1976
Limbu	lif	ST, TB	van Driem 1987
Lisu	lis	ST, TB	Bradley 2003
Lotha Naga	njh	ST, TB	Acharya 1975
Lushai (Mizo)	lus	ST, TB	Burling 1957

Language	ISO	Class	Sources
Magahi	mag	Iir, IA	Verma, S. 2003
Magar, Eastern	mgp	ST, TB	Shepherd & Shepherd 1971
Maithili	mai	Iir, IA	Yadav 1996, 2003
Malayalam	mal	Dr	Asher & Kumari 1997
Malto	kmj, mjt	Dr	Mahapatra 1979; Steever 1998b
Manangba (Manange)	mmm	ST, TB	Hildebrandt 2004
Mao Naga	nbi	ST, TB	Giridhar 1994
Marathi	mar	Iir, IA	Pandharipande 1997, 2003
Maria, Hill (Abujhmaria)	mrr	Dr	Natarajan 1985
Maria, Dandami	daq	Dr	Soundararaj & Soundararaj 1999
Marwari	rwr	Iir, IA	Gusain 2004
Meithei (Manipuri)	mni	ST, TB	Chelliah 1997
Mewati	wtm	Iir, IA	Gusain 2003
Mishmi, Digaro	mhu	ST, TB	Sastry 1984
Mising (Miri)	mrq	ST, TB	Prasad 1991
Mundari	unr	AA, Mu	Osada 2008
Naga, Phom	nph	ST, TB	Burling & Phom 1998
Nar Phu	npa	ST, TB	Noonan 2003b
Nepali	nep	Iir, IA	Riccardi 2003
Newar, Dolakha	new	ST, TB	Genetti 2007
Newar, Kathmandu	new	ST, TB	Hargreaves 2003
Nicobarese, Car	caq	AA, Ni	Das 1977
Nicobarese, Central	ncb	AA, Ni	Radhakrishnan 1981
Onge	oon	An	Abbi 2006
Oriya	ori	Iir, IA	Ray 2003
Palula	phl	Iir, IA	Liljegren 2016
Paniya	pcg	Dr	Daniel & Stephen 2003
Panjabi, Eastern	pan	Iir, IA	Bhatia 1993; Malik 1995
Pardhi (Bahelia)	pcl	Iir, IA	Srivastava 1968
Parenga (Gorum)	pcj	AA, Mu	Anderson & Rau 2008
Parji (Duruwa)	pci	Dr	Burrow & Bhattacharya 1953
Pashto, Northeastern	pbu	Iir, Ir	Elfenbein 1997c
Pashto, Southeastern	pbt	Iir, Ir	Elfenbein 1997c
Pengo	peg	Dr	Burrow & Bhattacharya 1970
Phake	phk	TK	Morey 2005
Pnar	pbv	AA, Kh	Ring 2015
Purik	prx	ST, TB	Zemp 2014
Rabha	rah	ST, TB	Joseph 2007

Language	ISO	Class	Sources
Rājbanshi	rjs	Iir, IA	Wilde 2008
Remo (Bonda)	bfw	AA, Mu	Anderson & Harrison 2008a
Rongmei Naga	nbu	ST, TB	Sreedhar 1979
Sadri (Sadani)	sck	Iir, IA	Jordan-Horstmann 1969
Santali	sat	AA, Mu	Ghosh 2008
Saurashtra	saz	Iir, IA	Norihiko 1991
Sawi	sdg	Iir, IA	Bashir 2003
Seke, Tangbe	skj	ST, TB	Honda 2003
Sema (Sumi Naga)	nsm	ST, TB	Sreedhar 1976
Seraiki	skr	Iir, IA	Shackle 1976
Shekawati	swv	Iir, IA	Gusain 2001
Sherpa	xsr	ST, TB	Kelly 2004
Shina, Gilgit	scl	Iir, IA	Radloff 1999
Shina, Kohistani	plk	Iir, IA	Schmidt & Kohistani 2008
Sikkimese (Denjongka)	sip	ST, TB	Yliniemi 2005
Sindhi	snd	Iir, IA	Khubchandani 2003; Nihilani 1999
Sinhalese	sin	Iir, IA	Gair 2003
Sora	srb	AA, Mu	Anderson & Harrison 2008b
Spiti	spt	ST, TB	Sharma, S. R. 1979
Sunwar	suz	ST, TB	Borchers 2008
Tamang, Eastern	taj	ST, TB	Mazaudon 2003
Tamil, Kanniyakumari	tam	Dr	Christdas 1988
Tamil, Madras	tam	Dr	Annamalai & Steever 1998; Keane 2004
Tangkhul Naga	nmf	ST, TB	Arokianathan 1987
Telugu	tel	Dr	Krishnamurti 1998
Thado Chin	tcz	ST, TB	Thirumalai 1972; Haokip 2008
Thakali	ths	ST, TB	Hari 1970
Thangmi	thf	ST, TB	Turin 2004
Tharu, Chitwan	the	Iir, IA	Boehm 1998
Tharu, Danguara	thl	Iir, IA	Boehm 1998
Tharu, Kochila	thq	Iir, IA	Boehm 1998
Tharu, Rana	thr	Iir, IA	Boehm 1998
Thulung Rai	tdh	ST, TB	Lahaussois 2003
Toda	tcx	Dr	Sakthivel 1976
Torwali	trw	Iir, IA	Lunsford 2001; Bashir 2003
Tshangla	tsj	ST, TB	Andvik 2003
Tulu	tey	Dr	Bhat 1998
Urali	url	Dr	Lal 1991

Language	ISO	Class	Sources
Urdu	urd	Iir, IA	Schmidt 2003
Vaagri Boli	vaa	Iir, IA	Varma 1970
Wakhi	wbl	Iir, Ir	Bashir 2009; Satoko 2005
Wambule	wme	ST, TB	Opgenort 2004
Yerava (Ravula)	yea	Dr	Mallikarjun 1993
Yerukala	yeu	Dr	Varma 1978b
Zanskari	zau	ST, TB	Hoshi & Tsering 1978; Jerry & Jerry 2013

Appendix B: Languages sampled outside of South Asia

This section lists 55 language varieties that are spoken in the regions surrounding South Asia (e.g., Southeast Asia, East Asia, Central Asia, etc.). These languages are included, along with those from section 1 (above), on the maps that accompany the paper *Retroflexion in South Asia: Typological, genetic, and areal patterns*.

Language	ISO	Class	Sources
Akha	ahk	ST, TB	Lewis 1968
Amharic	amh	Semitic	Hayward & Hayward 1999
Arabic, Gulf	afb	Semitic	Qafisheh 1977
Arabic, San'ani	ayn	Semitic	Watson 2002
Ashkun	ask	Iir, Nu	Strand 2011d
Bai, Yunnan	bfs	ST, TB	Wiersma 2003
Burmese	mya	ST, TB	Watkins 2001; Wheatley 2003
Cantonese	yue	ST, Ch	Zee 1999; Bauer & Matthews 2003
Caodeng Rgyalrong	jya	ST, TB	Sun 2003
Cone (Choni)	cda	ST, TB	Jacques 2014
Dari	prs	Iir, Ir	Rees 2008
Dulong	duu	ST, TB	LaPolla 2003a
Harsusi	hss	Semitic	Johnstone 1977
Jiarong (Cogtse)	jya	ST, TB	Nagano 2003
Jinghpö	kac	ST, TB	Qingxia & Diehl 2003
Kamviri	xvi	Iir, Nu	Strand 2011a.
Karen, Pwo	kjp	ST, TB	Kato 2003
Kati	bsh	Iir, Nu	Strand 2011b
Kayah, Eastern	eky	ST, TB	Solnit 2003
Khmer (Cambodian)	khm	AA, Khmeric	Bisang 2014
Kirghiz	kir	Turkic	Kirchner 1998
Koroshi	ktl	Iir, Ir	Nourzaei et. al. 2015

Language	ISO	Class	Sources
Lahu	lhu	ST, TB	Matisoff 2003
Lao	lao	TK	Enfield 2007
Mandaic	mid	Semitic	Malone 1997
Mandarin, Standard	cmn	ST, Ch	Duanmu 2002
Oirat	xal	Mongolic	Birtalan 2003
Palaung, Shwe	pll	AA, Pa	Mak 2012
Parachi	prc	Iir, Ir	Kieffer 2009
Pashto, Northwest	pbu	Iir, Ir	Elfenbein 1997c
Pashto, Southwest	pbt	Iir, Ir	Elfenbein 1997c
Persian (Farsi)	pes	Iir, Ir	Majidi & Ternes 1999; Windfuhr 1997
Pumi, Dayang	pmj	ST, TB	Matisoff 1997
Pumi, Niuwozi	pmj	ST, TB	Ding 2003
Pumi, Wadu	pmi	ST, TB	Daudey 2014
Qiang	cng	ST, TB	LaPolla 2003b
Rawang	raw	ST, TB	Morse 1963
Shughni	sgn	Iir, Ir	Edelman & Dodykhudoeva 2009
Somali	som	Cushitic	Saeed 1999
Tajiki	tgk	Iir, Ir	Windfuhr & Perry 2009
Thai	tha	TK	Tingsabadh & Abramson 1999
Thebo	cda (?)	ST, TB	Lin 2014
Tibetan, Amdo (Labrang)	adx	ST, TB	Makley et al. 1999
Tibetan, Amdo (Ndzorge)	adx	ST, TB	Sun 1986
Tibetan, Khams (Dege)	khg	ST, TB	Häsler 1999
Tibetan, Khams (Dongwang)	khg	ST, TB	Bartee 2014
Tibetan, Khams (Kami)	khg	ST, TB	Chirkova 2014
Tibetan, Khams (Rgyalhang)	khg	ST, TB	Hongladarom 1996
Tibetan, Lhasa	bod	ST, TB	DeLancey 2003; Denwood 1999
Turkmen	tuk	Turkic	Schönig 1998
Uyghur	uig	Turkic	Comrie 1997; Hahn 1998
Uzbek	uzn	Turkic	Boeschoten 1998
Vietnamese, Northern	vie	AA, Vietic	Brunelle 2014
Vietnamese, Southern	vie	AA, Vietic	Brunelle 2014
Waigali	wbk	Iir, Nu	Strand 2011c

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